Nonlinear Singularities at Mixed Hyperelastic Corners: Operator Blow-up, Creasing, and Structural Bifurcation

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Abstract

This paper investigates the coupling between geometric stress concentrations and nonlinear structural instabilities in compressible hyperelasticity, focusing on a corner subject to mixed Dirichlet–Neumann boundary conditions. Existing literature treats fracture, cavitation, and flat-surface creasing in isolation, leaving the mixed-boundary corner unresolved. We establish a dichotomy in the asymptotic behaviour of the linearised tangent operator, dictated by the loading direction and the volumetric energy penalty. In tension, a logarithmic volumetric penalty yields a benign singularity where the tangent operator remains strongly elliptic and Fredholm, preserving the standard Kondratiev edge exponents. Under compression, the local volume contraction drives an algebraic blow-up of the operator norm. We show that this breakdown is a structural bifurcation rather than a material loss of ellipticity. Using a tangent-cone blow-up reduction, we prove that a localised surface creasing instability preempts pointwise material collapse. An imperfect bifurcation analysis demonstrates that the quadratic energy coefficient vanishes by symmetry, whereas the subcritical cubic coefficient is tip-convergent yet explicitly dependent on the global stress-intensity factor, an outcome of the characteristic edge exponent satisfying $\alpha > 1/2$. The theoretical framework is supported by a scale-covariant Mellin–Chebyshev spectral formulation that resolves the singular exponents and buckling thresholds with exponential accuracy.

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1 Introduction

1.1 Existing literature

The literature on singularities in finite and nonlinear hyperelasticity is organised around three major phenomena.

- **Crack tips in tension** (Knowles and Sternberg, 1970s). The classical work on nonlinear singularities treats a 360° corner (a crack) pulled in pure tension. In a neo-Hookean material the asymptotic displacement field departs markedly from linear fracture mechanics; the analysis is, however, confined to traction-free boundaries and pure tensile opening.
- **Cavitation** (Ball, 1982). Under severe hydrostatic tension the hyperelastic equations admit singular weak solutions in which a hole opens spontaneously in the interior of a solid domain — a point singularity with $\boldsymbol{F} \to \infty$.

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- **Surface creasing** (Biot, 1963; Hong et al., 2009). A compressed flat half-space of neo-Hookean material becomes unstable at a critical stretch ($\lambda \approx 0.54$); this instability is now understood as a *crease*, a point singularity nucleating on an initially flat surface.

1.2 The gap

The existing nonlinear-singularity literature largely avoids the configuration of interest here: *a geometric corner (e.g. 90°) carrying mixed boundary conditions (Dirichlet meeting Neumann) under compression.* Each strand addresses only part of it:

- Linear theory (Kondratiev) handles the mixed-BC corner but ignores the large-deformation kinematics.
- Nonlinear theory (Knowles–Sternberg) handles large deformations but avoids the Dirichlet–Neumann condition and avoids compression.
- Instability theory handles compression but typically on flat domains or symmetric geometries, ignoring a pre-existing geometric corner singularity.

The open question this paper addresses is how an already-present stress concentration at a geometric Dirichlet–Neumann corner couples with a nonlinear structural instability (buckling/creasing), and how the resulting tangent operators can be bounded.

2 Problem formulation

2.1 Finite-strain elastostatics as energy minimisation

We consider a hyperelastic body occupying a reference d-dimensional domain $\Omega \subset \mathbb{R}^d$ with boundary $\partial\Omega = \overline{\Gamma_D} \cup \overline{\Gamma_N}$, $\Gamma_D \cap \Gamma_N = \emptyset$. The deformation of the body is described by the displacement field $\boldsymbol{u} : \Omega \to \mathbb{R}^d$, the deformation map $\boldsymbol{y}(\boldsymbol{x}) = \boldsymbol{x} + \boldsymbol{u}(\boldsymbol{x})$, and the associated deformation gradient

$$\boldsymbol{F} = \nabla \boldsymbol{y} = \boldsymbol{I} + \nabla \boldsymbol{u}, \qquad J = \det \boldsymbol{F} > 0, \tag{1}$$

where ∇ denotes the gradient with respect to the reference configuration. The mechanical response is encoded in a strain energy density $\Psi(\boldsymbol{F})$, and the equilibrium configuration is the minimiser of the total potential energy

$$\mathcal{E}(\boldsymbol{u}) = \int_\Omega \Psi(\boldsymbol{F})\, dV - \int_\Omega \boldsymbol{f}_0 \cdot \boldsymbol{u}\, dV - \int_{\Gamma_N} \boldsymbol{t}_0 \cdot \boldsymbol{u}\, dS, \tag{2}$$

over the space of admissible displacements $U = \{\boldsymbol{u} \in H^1(\Omega; \mathbb{R}^d) : \boldsymbol{u} = \boldsymbol{g} \text{ on } \Gamma_D\}$, where $\boldsymbol{f}_0$ is a body force, $\boldsymbol{t}_0$ a prescribed traction on the Neumann boundary, and $\boldsymbol{g}$ the prescribed boundary displacement. We seek $\boldsymbol{u} \in U$ such that $\mathcal{E}(\boldsymbol{u}) \leq \mathcal{E}(\boldsymbol{v})$ for all admissible $\boldsymbol{v}$.

2.2 Stationarity, stress, and tangent modulus

A minimiser satisfies the stationarity condition $D_{\delta \boldsymbol{u}}\mathcal{E}(\boldsymbol{u}) = 0$ for all admissible variations $\delta\boldsymbol{u}$ (vanishing on Γ_D), which is the weak form of equilibrium

$$\int_\Omega \boldsymbol{P}(\boldsymbol{F}) : \nabla \delta\boldsymbol{u}\, dV = \int_\Omega \boldsymbol{f}_0 \cdot \delta\boldsymbol{u}\, dV + \int_{\Gamma_N} \boldsymbol{t}_0 \cdot \delta\boldsymbol{u}\, dS, \tag{3}$$

where the first Piola–Kirchhoff stress is the first derivative of the energy with respect to the deformation gradient,

$$\boldsymbol{P}(\boldsymbol{F}) = \frac{\partial \Psi}{\partial \boldsymbol{F}}. \tag{4}$$

The natural boundary condition associated with Γ_N is $\boldsymbol{P}\boldsymbol{N} = \boldsymbol{t}_0$, with $\boldsymbol{N}$ the reference outward unit normal. Because (3) is nonlinear in $\boldsymbol{u}$, it is solved iteratively by Newton's method. Linearising the residual about a current iterate $\bar{\boldsymbol{u}}$ in a direction $\Delta\boldsymbol{u}$ introduces the fourth-order tangent (elasticity) modulus $\mathbb{L}$, with components L_{iJkL},

$$\mathbb{L} = \frac{\partial \boldsymbol{P}}{\partial \boldsymbol{F}} = \frac{\partial^2 \Psi}{\partial \boldsymbol{F} \otimes \partial \boldsymbol{F}}, \qquad L_{iJkL} = \frac{\partial^2 \Psi}{\partial F_{iJ}\, \partial F_{kL}}, \tag{5}$$

which possesses the major symmetry $L_{iJkL} = L_{kLiJ}$, so that each Newton step requires solving the linear problem

$$\int_\Omega \nabla \delta\boldsymbol{u} : \mathbb{L}(\bar{\boldsymbol{F}}) : \nabla \Delta\boldsymbol{u}\, dV = -\, R(\bar{\boldsymbol{u}}; \delta\boldsymbol{u}), \tag{6}$$

with $R(\bar{\boldsymbol{u}}; \delta\boldsymbol{u})$ the residual of (3) evaluated at $\bar{\boldsymbol{u}}$. The well-posedness and conditioning of (6)—and hence the convergence of the whole nonlinear solver—are governed by the coercivity and boundedness of the bilinear form induced by $\mathbb{L}(\bar{\boldsymbol{F}})$. The central object of this work is therefore the behaviour of $\mathbb{L}$ as a function of the local deformation state.

2.3 The neo-Hookean material

To fix ideas and to exhibit explicit constants we use, as a running example, the compressible neo-Hookean strain energy density adopted in the cut-finite-element study [9] that motivates this analysis,

$$\Psi(\boldsymbol{F}) = \frac{\mu_e}{2}\left(\mathrm{tr}(\boldsymbol{F}^T\boldsymbol{F}) - d\right) + f(J), \tag{7}$$

where $\mu_e > 0$ is the shear modulus and $f(J)$ is a strictly convex volumetric penalty with $f(1) = 0$, $f'(1) = -\mu_e$, and $f(J) \to \infty$ as $J \to 0^+$. Two standard choices are

$$f_{\log}(J) = -\mu_e \log J + \lambda_e \log^2 J, \qquad f_{\mathrm{quad}}(J) = \tfrac{\kappa}{2}(J-1)^2, \tag{8}$$

with λ_e the Lamé constant and κ a bulk modulus; (7) with $f_{\log}$ reproduces the model of [9]. The corresponding stress and tangent are

$$\boldsymbol{P}(\boldsymbol{F}) = \mu_e \boldsymbol{F} + f'(J)\, J\, \boldsymbol{F}^{-T}, \tag{9}$$

$$L_{iJkL} = \mu_e\, \delta_{ik}\delta_{JL} + \left(f''(J)J^2 + f'(J)J\right) F^{-T}_{iJ} F^{-T}_{kL} - f'(J)J\, F^{-T}_{iL} F^{-T}_{kJ}. \tag{10}$$

The first term $\mu_e\, \delta_{ik}\delta_{JL}$ is constant; the entire nonlinear part is controlled by J and the *inverse* deformation gradient $\boldsymbol{F}^{-1}$, yielding the structural bound

$$\|\mathbb{L}(\boldsymbol{F})\| \le \mu_e + \Theta(J)\, \|\boldsymbol{F}^{-1}\|^2, \tag{11}$$

where $\|\boldsymbol{F}^{-1}\| = \lambda_{\min}^{-1}$ is set by the smallest principal stretch and Θ is a scalar growth function fixed by f (computed explicitly in §3.4). Consequently $\mathbb{L}$ can become unbounded through volumetric collapse, $\lambda_{\min} \to 0$ (equivalently $\boldsymbol{F}^{-1} \to \infty$); whether a purely tensile blow-up of $\boldsymbol{F}$ (with $\boldsymbol{F}^{-1}$ bounded) leaves $\mathbb{L}$ finite depends on the growth of $\Theta(J)$, which turns out to discriminate sharply between the two penalties in (8). This dichotomy is the analytical backbone of the present paper.

2.4 The mixed-corner singularity

The configuration of interest is a right-angle junction where the two boundary-condition types meet: a Dirichlet face (Γ_D, prescribed displacement) abutting a traction-free Neumann face (Γ_N). In $\mathbb{R}^d$ this junction is a codimension-two *edge* Σ (a point for $d = 2$, a curve or surface for $d \geq 3$); its transverse geometry, in the two-plane normal to Σ, is the right-angle corner we analyse, and we use polar coordinates (r, θ) in that transverse plane. Such junctions arise generically wherever a clamped face meets a traction-free face — for instance along the base edges of a compressed body, the concrete setting taken up in Section 5. Classical Kondratiev/edge theory for the linearised problem predicts a transverse singularity

$$\boldsymbol{u}(r,\theta) \sim K\, r^{\alpha}\, \boldsymbol{\Phi}(\theta), \qquad \alpha \in (0,1), \tag{12}$$

with r the distance to the edge Σ, K a generalised stress-intensity factor and α the (geometry- and Poisson-ratio-dependent) singularity exponent. The deformation gradient then scales as $\boldsymbol{F} = \boldsymbol{I} + \mathcal{O}(Kr^{\alpha-1})$, so $\|\boldsymbol{F}\| \to \infty$ as $r \to 0$.

The question this raises—and which the remainder of the paper addresses—is how this kinematic singularity propagates into the nonlinear tangent operator (5). In view of (11), the answer hinges on which principal stretch diverges: a tensile corner keeps $\boldsymbol{F}^{-1}$ bounded and (for a suitable penalty) leaves $\mathbb{L}$ well behaved, whereas a compressive corner drives $\lambda_{\min} \to 0$ and blows up the *norm* of $\mathbb{L}$ (severe ill-conditioning). The tangent nonetheless remains strongly elliptic — the compressive instability is *structural* (a corner analogue of creasing), not a change of type of the PDE. The following sections analyse the two scenarios in turn (Section 3) and then show how a small imperfection regularises the compressive case through a structural bifurcation (Section 4).

3 Asymptotic behaviour of the tangent operator at a mixed corner

3.1 Geometry and linearised kinematics

Let the reference domain $\Omega \subset \mathbb{R}^d$ ($d \geq 2$) carry a right-angle Dirichlet–Neumann edge Σ of codimension two. Near a point of Σ split $\mathbb{R}^d = T\Sigma \oplus \Pi$ into the edge tangent space and the normal two-plane Π, and use polar coordinates (r, θ) in Π with $r \in [0, R]$, $\theta \in [0, \pi/2]$, where $r = \mathrm{dist}(\cdot, \Sigma)$. Write B_ρ for the tubular neighbourhood $\{\boldsymbol{x} \in \Omega : \mathrm{dist}(\boldsymbol{x}, \Sigma) < \rho\}$ of the edge. The leading singularity is governed by the transverse problem in Π, modulated smoothly along Σ; for $d = 2$ the edge degenerates to a point and B_ρ to the corner disc.

- **Dirichlet boundary (Γ_D):** $\theta = 0$, deformation prescribed, $\boldsymbol{y}(r, 0) = \boldsymbol{g}(r)$.
- **Neumann boundary (Γ_N):** $\theta = \pi/2$, traction-free, $\boldsymbol{P}\boldsymbol{N} = \boldsymbol{0}$.

Classical Kondratiev theory for the *linearised* problem gives the leading-order singular displacement

$$\boldsymbol{u}(r,\theta) \sim K\, r^{\alpha}\, \boldsymbol{\Phi}(\theta), \tag{13}$$

with K the generalised stress-intensity factor (set by the far field), $\boldsymbol{\Phi}$ a smooth angular profile, and $\alpha \in (0, 1)$ the geometry- and Poisson-dependent exponent. Consequently

$$\boldsymbol{F} = \boldsymbol{I} + \mathcal{O}\big(K\, r^{\alpha-1}\big)\,, \qquad \|\boldsymbol{F}\| \to \infty \text{ as } r \to 0. \tag{14}$$

Whether this kinematic blow-up contaminates the tangent operator $\mathbb{L} = \partial^2\Psi/\partial\boldsymbol{F}\,\partial\boldsymbol{F}$ depends not on $\|\boldsymbol{F}\|$ but on which principal stretches diverge, which is the content of the rest of this section.

3.2 Standing assumptions on the stored energy

The mechanism we describe is not specific to the neo-Hookean model; that model only serves to fix ideas and to exhibit explicit constants. We therefore phrase the analysis for a general frame-indifferent energy and isolate the structural properties that the argument actually uses. Throughout, $\|\cdot\|$ denotes the operator (spectral) norm on tensors, $0 < \lambda_{\min}(\boldsymbol{F}) \leq \cdots \leq \lambda_{\max}(\boldsymbol{F})$ are the singular values of $\boldsymbol{F}$ (the d principal stretches), and we recall $\|\boldsymbol{F}^{-1}\| = \lambda_{\min}(\boldsymbol{F})^{-1}$.

Assumption 1 (Regularity and frame indifference)**.** The stored energy $\Psi : \mathrm{GL}^+(d) \to \mathbb{R}$ is objective, $\Psi \in C^2(\mathrm{GL}^+(d))$, and $\Psi(\boldsymbol{F}) \to +\infty$ as $\det \boldsymbol{F} \to 0^+$.

Assumption 2 (Inverse-channel growth of the tangent)**.** There exist a continuous, nondecreasing function $\Theta : (0,\infty) \to (0,\infty)$ and a constant $c_0 \geq 0$ such that the tangent modulus $\mathbb{L}(\boldsymbol{F}) = D^2\Psi(\boldsymbol{F})$ satisfies, for all $\boldsymbol{F} \in \mathrm{GL}^+(d)$,

$$\|\mathbb{L}(\boldsymbol{F})\| \;\leq\; c_0 \;+\; \Theta\big(\det \boldsymbol{F}\big)\, \|\boldsymbol{F}^{-1}\|^2. \tag{15}$$

Assumption 3 (Uniform strong ellipticity on bounded-distortion sets)**.** For every $0 < c \leq 1$ and $M \geq 1$ there is $\gamma(c, M) > 0$ such that whenever $\lambda_{\min}(\boldsymbol{F}) \geq c$ and $\det \boldsymbol{F} \leq M$,

$$L_{iJkL}(\boldsymbol{F})\, m_i N_J\, m_k N_L \;\geq\; \gamma(c, M)\, |\boldsymbol{m}|^2\, |\boldsymbol{N}|^2 \qquad \forall\, \boldsymbol{m}, \boldsymbol{N} \in \mathbb{R}^2. \tag{16}$$

The decisive feature of Assumption 2 is that the inverse-stretch factor $\|\boldsymbol{F}^{-1}\|^2 = \lambda_{\min}^{-2}$ is the *only* algebraic channel through which the Hessian can blow up; the dependence on $\|\boldsymbol{F}\|$ (equivalently on $\lambda_{\max}$) enters solely through the scalar $\Theta(\det \boldsymbol{F})$. Whether a tensile corner ($\lambda_{\max} \to \infty$, $\det \boldsymbol{F} \to \infty$, $\lambda_{\min}$ bounded below) is benign therefore reduces to the *growth rate* of Θ, which we quantify in §3.4. Materials with genuine tensile stiffening of the deviatoric part (Gent, Arruda–Boyce), whose tangent diverges through $\lambda_{\max}$ directly rather than through $\Theta(\det \boldsymbol{F})$, violate (15) and are excluded.

Lemma 1 (Assumptions for the neo-Hookean model)**.** *Let Ψ be the neo-Hookean energy* (7) *with $\mu_e > 0$ and $f \in C^2(0,\infty)$. Then Assumption 2 holds with $c_0 = \mu_e$ and*

$$\Theta(J) = \big|\, f''(J)J^2 + f'(J)J \,\big| + \big|\, f'(J)J \,\big|, \tag{17}$$

and Assumption 3 holds on the convexity set $\{f''(J) \geq 0\}$ — in particular on all of compression $J \leq 1$ and moderate tension for $f_{\log}$, and globally for f_{quad} ($f'' = \kappa > 0$); it can fail under extreme dilation $J > J_*$ for $f_{\log}$ (Remark 1). Assumption 1 additionally requires the barrier property $f(J) \to \infty$ as $J \to 0^+$: this holds for $f_{\log}$ but fails *for f_{quad}, since $f_{\text{quad}}(J) \to \frac{\kappa}{2} < \infty$ (see Remark 4 and Scenario B).*

Proof. Objectivity and C^2 regularity on $\mathrm{GL}^+(d)$ are immediate; the barrier hypothesis $f(J) \to \infty$ as $J \to 0^+$, when assumed, gives the blow-up required by Assumption 1. In (10) the first term is the scaled identity, of norm μ_e. Each of the remaining two terms is a rank-one, respectively transposition, contraction of $\boldsymbol{F}^{-T} \otimes \boldsymbol{F}^{-T}$; since $\|\boldsymbol{F}^{-T}\| = \|\boldsymbol{F}^{-1}\|$, each is bounded in operator norm by its scalar prefactor times $\|\boldsymbol{F}^{-1}\|^2$. The triangle inequality gives (15) with $c_0 = \mu_e$ and Θ as in (17). For Assumption 3, contracting (10) with $\boldsymbol{m} \otimes \boldsymbol{N}$ and writing $\xi := \boldsymbol{m} \cdot \boldsymbol{F}^{-T} \boldsymbol{N} = \boldsymbol{N} \cdot \boldsymbol{F}^{-1} \boldsymbol{m}$, the two volumetric terms combine ($(f''J^2 + f'J)\xi^2 - f'J\,\xi^2$) to the single clean form

$$L_{iJkL} m_i N_J m_k N_L = \mu_e\, |\boldsymbol{m}|^2 |\boldsymbol{N}|^2 + f''(J)\, J^2\, \xi^2. \tag{18}$$

Wherever $f''(J) \geq 0$ the second term is nonnegative and (16) holds with $\gamma = \mu_e$, uniformly in $\boldsymbol{F}$ (no distortion bound needed); this covers f_{quad} ($f'' = \kappa > 0$) everywhere and $f_{\log}$ on $\{f''_{\log} \geq 0\}$. □ □

Remark 1 (Strength and a tension caveat). Identity (18) is sharper than (16): the neo-Hookean tangent is strongly elliptic with constant *exactly* μ_e wherever $f''(J) \geq 0$, and the volumetric term only adds to it. The Hessian of the quadratic (Hookean) part is constant, which is why the *deviatoric* response never blows up; only the *volumetric* response, through Θ, can. One caveat for $f_{\log}$: since $f''_{\log}(J) = (\mu_e + 2\lambda_e - 2\lambda_e \log J)/J^2$, convexity is lost once $J > J_* := \exp\bigl(1 + \mu_e/(2\lambda_e)\bigr)$, and then (18) can turn negative for ξ near its maximal value $|\boldsymbol{m}||\boldsymbol{N}|/\lambda_{\min}$. Hence Assumption 3 holds throughout *compression* ($J \leq 1$) and moderate tension, but $f_{\log}$ can lose strong ellipticity under *extreme dilation* ($J > J_*$)—a tension-side phenomenon distinct from the compressive analysis, and one more reason the bounded-J qualifier matters in §3.5. The globally convex f_{quad} is free of this.

3.3 The non-compressive hypothesis

The qualitative split between the two scenarios is captured by a single quantity: the smallest principal stretch near the corner. We call the corner singularity *non-compressive* when $\lambda_{\min}$ stays bounded away from zero, even though $\lambda_{\max}$ (hence $\|\boldsymbol{F}\|$ and $\det \boldsymbol{F}$) may diverge.

Assumption 4 (Non-compressive singularity, (NC))**.** There exist $\rho \in (0, R]$ and $c \in (0, 1]$ such that the equilibrium deformation satisfies

$$\lambda_{\min}(\boldsymbol{F}(\boldsymbol{x})) \geq c, \quad \text{equivalently} \quad \|\boldsymbol{F}^{-1}(\boldsymbol{x})\| \leq c^{-1}, \qquad \text{for a.e. } \boldsymbol{x} \in B_\rho. \tag{19}$$

We stress what (19) does *not* say. It does not require incompressibility, and crucially it does *not* bound $\det \boldsymbol{F}$: in a tensile corner $\lambda_{\max} \sim r^{\alpha-1} \to \infty$, so $J = \det \boldsymbol{F} = \prod_i \lambda_i \sim r^{\alpha-1} \to \infty$ as well (one diverging stretch, the remaining $d-1$ bounded). The only quantity (NC) controls is $\|\boldsymbol{F}^{-1}\| \leq c^{-1}$. Physically, (NC) states that the material is pulled around the corner (tension/shear) rather than crushed into it. Because $\det \boldsymbol{F}$ is *not* bounded under (NC), the factor $\Theta(\det \boldsymbol{F})$ in (15) need not stay bounded; the next subsection makes its growth explicit, and this is precisely where the two volumetric penalties (8) part ways. Scenario B (§3.6) treats the complementary, compressive case in which (19) fails.

3.4 Explicit volumetric growth functions

The bound (15) exposes *two independent channels* through which the tangent can diverge, and they are kinematically decoupled:

$$\|\mathbb{L}(\boldsymbol{F})\| \leq \mu_e + \underbrace{\Theta(\det \boldsymbol{F})}_{\substack{\text{dilation channel} \\ J\to\infty}} \cdot \underbrace{\|\boldsymbol{F}^{-1}\|^2}_{\substack{\text{inverse channel} \\ \lambda_{\min}\to 0}}. \tag{20}$$

The *inverse channel* is the one usually blamed for trouble: it requires $\lambda_{\min} \to 0$, i.e. compression to zero volume, and is the subject of Scenario B. The *dilation channel*, $\Theta(\det \boldsymbol{F}) \to \infty$ as $J \to \infty$, is activated by *expansion* and is precisely what a tensile corner switches on, since $\lambda_{\max} \sim r^{\alpha-1} \to \infty$ forces $J \to \infty$ even with $\lambda_{\min}$ bounded below by (NC). Whether this matters is therefore not a property of the corner or of the sign of the loading, but of the *growth rate of* Θ—a property of the volumetric penalty f alone. We now evaluate it for the two penalties (8); the contrast shows that "tension is benign" is a model-dependent statement.

Logarithmic penalty. For $f_{\log}(J) = -\mu_e \log J + \lambda_e \log^2 J$,

$$f'(J)J = -\mu_e + 2\lambda_e \log J, \qquad f''(J)J^2 = \mu_e + 2\lambda_e - 2\lambda_e \log J, \tag{21}$$

so that the combination collapses to a constant,

$$f''(J)J^2 + f'(J)J = 2\lambda_e, \tag{22}$$

and therefore

$$\Theta_{\log}(J) = 2\lambda_e + \big|\, 2\lambda_e \log J - \mu_e \,\big| = \mathcal{O}\big(\log J\big) \quad \text{as } J \to \infty. \tag{23}$$

The volumetric stiffening grows only *logarithmically* in J.

Quadratic penalty. For $f_{\text{quad}}(J) = \frac{\kappa}{2}(J-1)^2$,

$$f'(J)J = \kappa J(J-1), \qquad f''(J)J^2 = \kappa J^2, \qquad f''(J)J^2 + f'(J)J = \kappa J(2J-1), \tag{24}$$

so

$$\Theta_{\text{quad}}(J) = \kappa J\,|2J-1| + \kappa J\,|J-1| = \mathcal{O}\big(\kappa J^2\big) \quad \text{as } J \to \infty. \tag{25}$$

The volumetric stiffening grows *quadratically* in J.

Consequence at the corner. We combine (15) with (NC) ($\|\boldsymbol{F}^{-1}\|^2 \le c^{-2}$) and the tensile kinematics. In the non-compressive regime exactly one stretch diverges: $\lambda_{\max} \sim |K|\, r^{\alpha-1}$ while the remaining $d-1$ stretches stay bounded (in particular $\lambda_{\min}$ bounded below), so $J = \det \boldsymbol{F} = \prod_i \lambda_i \sim r^{\alpha-1}$. (If instead $\nabla \boldsymbol{u}$ had two diverging eigenvalues, the corresponding subdeterminant would give $J \sim r^{2(\alpha-1)}$; we work in the one-large-stretch regime characteristic of a Dirichlet–Neumann opening, where $J \sim r^{\alpha-1}$.) The tangent norm near the corner then obeys

$$\|\mathbb{L}(\boldsymbol{x})\| \le \mu_e + c^{-2}\,\Theta\big(\det \boldsymbol{F}(\boldsymbol{x})\big) \sim \begin{cases} \mu_e + \mathcal{O}\big(\log \frac{1}{r}\big), & f = f_{\log}, \\ \mathcal{O}\big(r^{-2(1-\alpha)}\big), & f = f_{\text{quad}}. \end{cases} \tag{26}$$

Thus the logarithmic penalty keeps the tangent bounded up to a logarithm, whereas the quadratic penalty produces a genuine *algebraic* blow-up of the tangent *even in pure tension.* This is the dilation channel of (20) acting alone, with the inverse channel inert ($\|\boldsymbol{F}^{-1}\|$ bounded); it is algebraically singular of a severity comparable to—though mechanistically distinct from—the compressive blow-up of Scenario B, which instead runs through $\|\boldsymbol{F}^{-1}\|^2$. The often-repeated statement that "tension is harmless for neo-Hookean" is therefore an artefact of the logarithmic volumetric model and is false for the quadratic one.

Physical reading: the large-dilation response. The dichotomy is, at bottom, a statement about how the bulk energy behaves under unbounded *expansion*, the loading a tensile corner imposes:

volumetric model	energy as $J \to \infty$	tangent $\Theta(J)$
$\lambda_e \log^2 J$	$\sim \log^2 J$ (soft)	$\mathcal{O}(\log J)$
$\frac{\kappa}{2}(J-1)^2$	$\sim J^2$ (stiff)	$\mathcal{O}(J^2)$

A penalty that is polynomial in J makes the material resist further dilation ever more stiffly, and the linearised modulus faithfully reports that stiffness as a blow-up; the logarithmic penalty is gentle about expansion and does not. Both penalties are designed to diverge correctly as $J \to 0$ (resisting collapse, the inverse channel); their large-J behaviour, which the dilation channel probes, was never physically constrained, and that latitude is exactly where they part. The exact cancellation (22) is the analytic signature of this softness. We make the benign case precise in Theorem 1 and flag the algebraic case in Remark 4.

3.5 Scenario A: the tensile (non-compressive) singularity

We work under the mild-growth condition isolated above.

Assumption 5 (Mild volumetric growth)**.** The growth function satisfies $\Theta(J) = \mathcal{O}\big((\log J)^p\big)$ as $J \to \infty$ for some $p \geq 0$.

By (23), $f_{\log}$ satisfies Assumption 5 with $p = 1$; by (25), f_{quad} does not.

Lemma 2 (Subordinate tangent)**.** *Under Assumptions 1, 2, 5 and (NC), for every $\varepsilon > 0$ there is C_ε with*

$$\|\mathbb{L}(\boldsymbol{x})\| \leq C_\varepsilon\, r^{-\varepsilon} \qquad \text{for a.e. } \boldsymbol{x} \in B_\rho. \tag{27}$$

In particular $\mathbb{L} \in L^q(B_\rho)$ for every $q < \infty$, and $\|\mathbb{L}\|$ is dominated by any negative power of r.

Proof. By (15) and (NC), $\|\mathbb{L}\| \leq \mu_e + c^{-2}\Theta(\det \boldsymbol{F})$. Assumption 5 and $\det \boldsymbol{F} = \mathcal{O}(r^{\alpha-1})$ give $\Theta(\det \boldsymbol{F}) = \mathcal{O}\big((\log \frac{1}{r})^p\big)$, and $(\log \frac{1}{r})^p \leq C_\varepsilon r^{-\varepsilon}$ on $(0, \rho]$ for every $\varepsilon > 0$. □ □

Lemma 3 (Strong ellipticity outside the dilation core)**.** *Let $f = f_{\log}$ and let $J_* = \exp\big(1+\mu_e/(2\lambda_e)\big)$ be the dilation threshold of Remark 1, beyond which $f''_{\log} < 0$. Under Assumptions 1, 3 and (NC), with the tensile kinematics $J \sim |K|\, r^{\alpha-1}$ of §3.4, define the* dilation core radius

$$r_* := \big(J_*/|K|\big)^{1/(\alpha-1)} = \big(|K|/J_*\big)^{1/(1-\alpha)}, \tag{28}$$

so that $J(\boldsymbol{x}) \leq J_*$ for $r \geq r_*$. Then on the annular region $A_* := \{\boldsymbol{x} \in B_\rho : r_* < r < \rho\}$ the tangent satisfies the Legendre–Hadamard condition with the constant *ellipticity modulus μ_e,*

$$L_{iJkL}(\boldsymbol{x})\, m_i N_J m_k N_L \;\geq\; \mu_e\, |\boldsymbol{m}|^2 |\boldsymbol{N}|^2 \qquad \text{for a.e. } \boldsymbol{x} \in A_*, \quad \forall \boldsymbol{m}, \boldsymbol{N} \in \mathbb{R}^2. \tag{29}$$

On the complementary core $\{r \leq r_*\}$, where $J > J_*$, the constant-modulus bound (29) *no longer holds, and strong ellipticity of $f_{\log}$* can *fail (Remark 1): wherever the sharper condition $|f''(J)|J^2\, \|\boldsymbol{F}^{-1}\|^2 > \mu_e$ is met, the directional modulus $\mu_e + f''(J)J^2\, \xi^2/(|\boldsymbol{m}|^2|\boldsymbol{N}|^2)$ turns negative for ξ near its maximum. We do not claim the loss is pointwise throughout the core — only that the core is where it can occur, since $f''(J)J^2 < 0$ there is necessary for it.*

Proof. By the exact rank-one identity (18),

$$L_{iJkL} m_i N_J m_k N_L = \mu_e\, |\boldsymbol{m}|^2 |\boldsymbol{N}|^2 + f''(J)\, J^2\, \xi^2, \qquad \xi := \boldsymbol{m} \cdot \boldsymbol{F}^{-T} \boldsymbol{N},$$

in which the volumetric prefactor is the single scalar $f''(J)J^2 = \mu_e + 2\lambda_e - 2\lambda_e \log J$ (cf. (22)). This is nonnegative precisely when $\log J \leq 1 + \mu_e/(2\lambda_e)$, i.e. $J \leq J_*$. Since $J \sim |K| r^{\alpha-1}$ is decreasing in r (recall $\alpha < 1$), the threshold $J = J_*$ is crossed at the single radius (28), and $J \leq J_*$ throughout A_*. There the second term only adds to μ_e, giving (29). For $r < r_*$ we have $J > J_*$, hence $f''(J)J^2 < 0$; taking $\boldsymbol{m}, \boldsymbol{N}$ aligned so that ξ^2 attains its maximum $|\boldsymbol{m}|^2 |\boldsymbol{N}|^2\, \|\boldsymbol{F}^{-1}\|^2$ (admissible since $\|\boldsymbol{F}^{-1}\| \leq c^{-1}$ is only an upper bound, with equality possible) drives the form below zero once $|f''(J)J^2|\, \|\boldsymbol{F}^{-1}\|^2 > \mu_e$. □ □

The picture is therefore sharper, and more honest, than "tension is harmless": for the logarithmic penalty the tensile corner is strongly elliptic on the annulus A_* outside a *dilation core* $\{r \leq r_*\}$, inside which $f_{\log}$ itself — not the corner — *may* cease to be strongly elliptic because the material is dilated past J_* (where the sharper condition $|f''(J)|J^2 \|\boldsymbol{F}^{-1}\|^2 > \mu_e$ binds). The core radius r_* in (28) shrinks as the threshold $J_* = \exp(1 + \mu_e/(2\lambda_e))$ grows, i.e. as the penalty is made stiffer against extreme dilation (large λ_e/μ_e); a model with $f'' \geq 0$ globally would have $r_* = 0$ and no core. This is the tensile mirror of the model-dependence already seen for Θ in §3.4, and it confines the genuinely delicate behaviour to a controlled inner region.

For the weighted well-posedness we recall two standard inequalities in the Kondratiev space $V^{1,2}_\beta(B_\rho)$—the closure of smooth functions vanishing on $\Gamma_D \cap \partial B_\rho$ under $\|\boldsymbol{v}\|^2_{V^{1,2}_\beta} = \int_{B_\rho} \big(r^{2\beta}|\nabla \boldsymbol{v}|^2 + r^{2\beta-2}|\boldsymbol{v}|^2\big)\, dV$.

Lemma 4 (Weighted Hardy and Korn inequalities [6, 8])**.** *Let $\beta \in \mathbb{R}$ avoid the (discrete) set of exceptional exponents of the corner. There are constants C_H, C_K depending only on β and the opening angle such that, for all $\boldsymbol{v} \in V^{1,2}_\beta(B_\rho)$ vanishing on $\Gamma_D \cap \partial B_\rho$,*

$$\int_{B_\rho} r^{2\beta-2}|\boldsymbol{v}|^2\, dV \le C_H \int_{B_\rho} r^{2\beta}|\nabla \boldsymbol{v}|^2\, dV, \tag{30}$$

$$\int_{B_\rho} r^{2\beta}|\nabla \boldsymbol{v}|^2\, dV \le C_K \int_{B_\rho} r^{2\beta}\, |\boldsymbol{\varepsilon}(\boldsymbol{v})|^2\, dV, \qquad \boldsymbol{\varepsilon}(\boldsymbol{v}) = \tfrac{1}{2}\big(\nabla \boldsymbol{v} + \nabla \boldsymbol{v}^T\big). \tag{31}$$

Let $\beta_\star$ be the weight exponent for which the *linear* elasticity corner problem at this Dirichlet–Neumann junction is Fredholm (equivalently $\beta_\star$ avoids the singular exponents α). The existence of such a weight, and the resulting Fredholm property in the weighted space $V^{1,2}_{\beta_\star}$, is the standard Kondratiev–Maz'ya–Nazarov–Plamenevsky theory for elliptic systems with conical/edge singularities [6, 8]: the operator is Fredholm precisely when $\beta_\star$ avoids the discrete spectrum of the corner operator pencil (the singular exponents), and the resolvent is then compact, so the spectrum is discrete and the critical mode $\boldsymbol{\phi}$ of Assumption 6 is genuinely attained. We may write the Newton-step bilinear form as

$$a(\Delta \boldsymbol{u}, \delta \boldsymbol{u}) := \int_{B_\rho} \nabla \delta \boldsymbol{u} : \mathbb{L}(\bar{\boldsymbol{F}}) : \nabla \Delta \boldsymbol{u}\, dV. \tag{32}$$

Theorem 1 (Well-posedness of the linearised step, tensile corner)**.** *Under Assumptions 1–5 and the non-compressive hypothesis (NC), the Newton-step form* (32) *is bounded and coercive modulo a compact perturbation: on any region Ω on which the tangent is strongly elliptic with the constant Legendre–Hadamard modulus μ_e (Lemma 3), there exist constants $\Lambda < \infty$, $\gamma_1 > 0$ and a compact form $k(\cdot,\cdot)$ such that, for all $\boldsymbol{u}, \boldsymbol{v} \in V^{1,2}_{\beta_\star}(\Omega)$,*

$$|a(\boldsymbol{u}, \boldsymbol{v})| \le \Lambda\, \|\boldsymbol{u}\|_{V^{1,2}_{\beta_\star}}\, \|\boldsymbol{v}\|_{V^{1,2}_{\beta_\star}}, \tag{33}$$

$$a(\boldsymbol{v}, \boldsymbol{v}) \ge \gamma_1\, \|\boldsymbol{v}\|^2_{V^{1,2}_{\beta_\star}} - k(\boldsymbol{v}, \boldsymbol{v}). \tag{34}$$

The consequences split according to whether the vertex lies in the elliptic region.

(A) Globally convex-volumetric barrier ($f'' \ge 0$ together with the standing Assumptions 1–5, e.g. $f_{\log}$ with a finite-extensibility cap $J_{\max} \le J_*$ on the dilation). *Then the dilation core is empty, $r_* = 0$, ellipticity holds on the* full *corner neighbourhood $\Omega = B_\rho$* including the vertex, *and* (33)–(34) *hold on $V^{1,2}_{\beta_\star}(B_\rho)$. Hence the linearised corner operator is Fredholm of index zero in $V^{1,2}_{\beta_\star}(B_\rho)$; for $\beta_\star$ avoiding the corner spectrum it is an isomorphism, its solutions admit the Kondratiev asymptotic expansion* (13) *at the vertex, and the governing singular exponents coincide with those of the linearised elasticity problem. This is the genuine corner result.*

(B) Logarithmic barrier $f_{\log}$ ($r_* > 0$). *The vertex sits inside the non-elliptic core $\{r \le r_*\}$ (Lemma 3), so the estimates hold only on the* vertex-free *annulus $\Omega = A_* = \{r_* < r < \rho\}$. There the operator is a regular uniformly elliptic problem:* (33)–(34) *give Fredholmness of index zero in $V^{1,2}_{\beta_\star}(A_*)$ and well-posedness on the annulus, but* no *corner-pencil asymptotics are asserted across the inner boundary $r = r_*$, which is a smooth interface carrying no vertex, not a conical point. The* *vertex behaviour of $f_{\log}$ is the open inner problem of Remark 2. In the mild-loading limit $r_* \downarrow 0$ (small $|K|$) the core shrinks below any fixed scale and the conclusion of (A) is recovered on every fixed sub-annulus down to mesh resolution.*

Proof. The bounds (33)–(34) are established on the elliptic region Ω ($= B_\rho$ in case (A), $= A_*$ in case (B)); the case split affects only whether the vertex is interior to Ω, hence whether the Kondratiev corner analysis applies.

Boundedness. By Cauchy–Schwarz and Lemma 2, on Ω the coefficient obeys the sharp bound $\|\mathbb{L}\| \le \mu_e + c^{-2}\Theta(J) = \mu_e + \mathcal{O}\big(\log \frac{1}{r}\big)$, a *logarithm*, not a power. Hence for every $\delta > 0$ there is C_δ with $\|\mathbb{L}\| \le C_\delta\, r^{-\delta}$, and

$$|a(\boldsymbol{u}, \boldsymbol{v})| \le \int_\Omega \|\mathbb{L}\|\, |\nabla \boldsymbol{u}|\, |\nabla \boldsymbol{v}|\, dV \le C_\delta \Big(\int_\Omega r^{2\beta_\star} |\nabla \boldsymbol{u}|^2\Big)^{1/2} \Big(\int_\Omega r^{-2\delta - 2\beta_\star} |\nabla \boldsymbol{v}|^2\Big)^{1/2}.$$

This exhibits a as bounded from $V^{1,2}_{\beta_\star} \times V^{1,2}_{-\beta_\star-\delta}$ (the dual pairing $V^{1,2}_{\beta_\star} \times V^{1,2}_{-\beta_\star}$ with a δ of weighted slack). The logarithm is subordinate to *every* algebraic weight, so the loss may be made smaller than the slack to the weight interval: let (β_-, β_+) be the maximal open interval of weights for which the *linear* corner pencil has no eigenvalue with $\Re$-part in $[\beta_\star - \delta, \beta_\star]$ (such a δ-band exists because $\beta_\star$ avoids the discrete corner spectrum, hence sits at a positive distance δ_0 from it). Choose $\delta < \delta_0$, so $\beta_\star - \delta$ lies in the same admissible interval as $\beta_\star$. In case (B) the domain $\Omega = A_*$ excludes the vertex, the two Kondratiev norms $V^{1,2}_{\beta_\star}$ and $V^{1,2}_{-\beta_\star-\delta}$ are then comparable (they differ only at the vertex, which A_* omits), and (33) follows directly with $\Lambda = \Lambda(\delta)$. In case (A) the vertex is interior to $\Omega = B_\rho$ and the two norms are *not* comparable there, so the same-space estimate is obtained by perturbation off the constant-coefficient symbol rather than by norm comparison. Write $\mathbb{L} = \mathbb{L}_{\text{lin}} + \mathbb{R}$, with $\mathbb{L}_{\text{lin}}$ the frozen linear-elasticity tensor at the vertex (whose pencil is Fredholm in $V^{1,2}_{\beta_\star}(B_\rho)$ for $\beta_\star$ off the corner spectrum, by Kondratiev–Maz'ya theory [6]) and $\mathbb{R} = \mathbb{L} - \mathbb{L}_{\text{lin}} = \mathcal{O}(\log \frac{1}{r})$ the nonlinear correction. The hypotheses of the relevant stability theorem are exactly: (a) $\mathbb{L}_{\text{lin}}$ is strongly elliptic and its pencil has no eigenvalue on $\Re\lambda = \beta_\star$; and (b) $\mathbb{R}$ is *subordinate* to the principal part, $\|\mathbb{R}\| = o(r^{-\delta})$ for every $\delta > 0$ (Lemma 2), hence of relative bound 0 with respect to the Hardy-weighted graph norm. Both are met here, so the standard subordinate-perturbation theorem of [6] applies: the admissible weight interval is unmoved and the *full* pair of estimates (33)–(34) — not boundedness alone — transfers from $\mathbb{L}_{\text{lin}}$ to $\mathbb{L}$ on $V^{1,2}_{\beta_\star}(B_\rho)$. In both cases the logarithmic growth does not shift the admissible interval; no singular weight is "absorbed" into $\boldsymbol{v}$.

Gårding inequality. By Lemma 3 the symbol $\mathbb{L}(\boldsymbol{x})$ is strongly elliptic on Ω with the constant Legendre–Hadamard modulus μ_e. Freezing the (measurable, uniformly elliptic) coefficients on a dyadic partition $\{r \sim 2^{-j}\} \cap \Omega$ and applying the constant-coefficient Gårding inequality on each annulus—valid because the principal symbol is positive definite on the Legendre–Hadamard cone—yields, after summation against the weight $r^{2\beta_\star}$,

$$a(\boldsymbol{v}, \boldsymbol{v}) \ge \mu_e\, c_K^{-1} \int_\Omega r^{2\beta_\star} |\nabla \boldsymbol{v}|^2\, dV - C \int_\Omega r^{2\beta_\star} |\boldsymbol{v}|^2\, dV,$$

where the step from Legendre–Hadamard positivity to coercivity is the classical one for strongly elliptic systems: strong ellipticity with modulus μ_e (Lemma 3) gives the constant-coefficient Gårding bound for the symmetrised gradient $\boldsymbol{\varepsilon}(\boldsymbol{v})$ on each frozen annulus, and the weighted Korn inequality (31) then converts control of $\boldsymbol{\varepsilon}(\boldsymbol{v})$ into control of the full gradient $\nabla \boldsymbol{v}$. The remainder term has a strictly higher power of r than the Hardy weight $r^{2\beta_\star - 2}$, hence by (30) it is relatively compact with respect to $\|\cdot\|_{V^{1,2}_{\beta_\star}}$; this is the compact form k, and we obtain (34) with $\gamma_1 = \mu_e\, c_K^{-1}$.

Fredholm property and exponents. Estimates (33)–(34) place the operator in the Kondratiev–Maz'ya framework [6]: it is Fredholm of index zero in $V^{1,2}_{\beta_\star}(\Omega)$ whenever $\beta_\star$ avoids the spectrum of the corner operator pencil. In case (A), where $\Omega = B_\rho$ contains the vertex, this is a genuine corner statement: because the nonlinear correction to the principal symbol is subordinate (it is $o(r^{-\delta})$ for every δ, by Lemma 2), the pencil coincides with that of linearised elasticity, and its eigenvalues are exactly the exponents α of (13), unchanged by the nonlinearity. In case (B), where $\Omega = A_*$ omits the vertex, the same estimates give Fredholmness and well-posedness on the annulus, but the inner boundary $r = r_*$ is a smooth interface, not a conical point, so the pencil eigenvalues describe no vertex asymptotics there; the corner expansion (13) is recovered only in the limit $r_* \downarrow 0$ on fixed sub-annuli, and the genuine vertex theory for $f_{\log}$ remains the open inner problem of Remark 2. $\square$ $\square$

Remark 2 (The dilation core). For $f_{\log}$ the conclusion of Theorem 1 is genuinely confined to A_*: the vertex itself sits inside the core $\{r \le r_*\}$, where $J > J_*$ and $f_{\log}$ is not strongly elliptic. Three readings are consistent with the analysis. (i) *Mild loading*: for stress-intensity $|K|$ small the core radius $r_* = (|K|/J_*)^{1/(1-\alpha)}$ is so small that the corner is elliptic on all of B_ρ down to mesh scale, and the theorem governs the entire computationally resolved region. (ii) *Sharper model statement*: the tensile corner "passes like linear elasticity" on A_*, and the only obstruction is the material law's own loss of ellipticity past J_*—a property of $f_{\log}$, not of the corner. (iii) *Open inner problem*: a complete vertex theory for $f_{\log}$ requires a separate analysis inside the non-elliptic core, which we leave open; a globally convex-volumetric barrier ($f'' \ge 0$, $r_* = 0$) removes the issue entirely and recovers the unrestricted statement. Physically, the loss of ellipticity past J_* is an artefact of the unbounded dilation that $f_{\log}$ permits: real polymer networks reach *finite extensibility*, and a volumetric law that caps the dilation, $J < J_{\max}$ (the bulk analogue of the Gent deviatoric model), with $J_{\max} \le J_*$ keeps $f'' > 0$ throughout the attainable range and so eliminates the core. The dilation core is thus a feature of the specific large-J tail of $f_{\log}$, not a physical catastrophe.

Remark 3 (Stress is singular, operator is not). The first Piola–Kirchhoff stress still blows up, $\boldsymbol{P} \sim \mu_e \boldsymbol{F} \to \infty$ as $r \to 0$, so the tensile corner is a genuine *stress* singularity. Theorem 1 asserts only that, on the annulus A_*, this singularity lives entirely in the *right-hand side* (the residual) of the Newton step, not in its operator: under a mild volumetric penalty the nonlinear problem inherits the regularity theory of the linear one, standard graded meshes restore optimal convergence, and no structural instability occurs. This is the precise sense in which a non-compressive corner "passes like linear elasticity" away from the dilation core.

Remark 4 (The quadratic penalty does *not* pass like linear elasticity). For f_{quad}, Assumption 5 fails: by (25) and (26) the tangent grows algebraically, $\|\mathbb{L}\| = \mathcal{O}(r^{-2(1-\alpha)})$, *even in pure tension.* The coefficients of (32) are then genuinely singular, of severity comparable to the compressive Scenario B, the boundedness step above fails for $\beta_\star$, and the admissible weight interval—hence the effective corner exponents seen by the nonlinear operator—shifts relative to linear elasticity. A sharp analysis of this case (the correct weight, and whether coercivity survives) is governed by the volumetric blow-up rather than by the deviatoric singularity and is left open here.

We emphasise that this serves as a *cautionary comparator* rather than a model we advocate: as written, $f'_{\text{quad}}(1) = 0 \neq -\mu_e$, so $\frac{\kappa}{2}(J-1)^2$ does not combine with the deviatoric term $\frac{\mu_e}{2}(\text{tr}(\boldsymbol{F}^T\boldsymbol{F}) - d)$ to leave the reference configuration stress-free, and in practice it is used together with an isochoric (deviatoric/volumetric) split of $\boldsymbol{F}$. The point is that a standard-looking polynomial penalty already forfeits the benign tensile behaviour enjoyed by the logarithmic model (8) actually used in [9].

Remark 5 (Self-consistency of (NC)). Hypothesis (NC) is compatible with the singular kinematics (14): an angular profile $\boldsymbol{\Phi}$ corresponding to opening/shear produces $\nabla \boldsymbol{u}$ whose dominant eigenvalue is positive (radial extension), so $\lambda_{\max} \to \infty$ while $\lambda_{\min} = \det \boldsymbol{F}/\lambda_{\max}$ need not approach zero. The complementary profile, in which the dominant eigenvalue of $\nabla \boldsymbol{u}$ is negative, drives $\lambda_{\min} \to 0$ and violates (NC); that is exactly Scenario B below.

3.6 Scenario B: the compressive singularity (the pathological case)

Now assume the far-field loading pushes the material *into* the corner ($K < 0$, radial compression).

Kinematic consequence. The naive linear prediction gives $\nabla \boldsymbol{u}$ a negative dominant eigenvalue scaling as $-|K|r^{\alpha-1}$. Because $\boldsymbol{F} = \boldsymbol{I} + \nabla \boldsymbol{u}$, as $r \to 0$ (or as the pseudo-time load factor t increases) there is a critical threshold where the local contraction annihilates the material volume:

$$\det \boldsymbol{F} \to 0 \implies J \to 0, \qquad \lambda_{\min} \to 0. \tag{35}$$

Operator blow-up (and its model dependence). Here it is the *inverse channel* of (20) that fires: $\|\boldsymbol{F}^{-1}\| = \lambda_{\min}^{-1} \to \infty$. The dilation channel $\Theta(J)$ is now probed at $J \to 0^+$ rather than $J \to \infty$, and—exactly as in tension, but with the roles reversed—its behaviour discriminates between the two penalties. Writing $J = \prod_i \lambda_i \sim \lambda_{\min}$ near the corner (one collapsing stretch, the remaining $d-1$ bounded, cf. §3.4),

$$\Theta_{\log}(J) \sim 2\lambda_e \left|\log J\right| \to \infty, \qquad \Theta_{\text{quad}}(J) \sim 2\kappa J \to 0 \qquad (J \to 0^+), \tag{36}$$

so that the tangent norm $\|\mathbb{L}\| \le \mu_e + \Theta(J)\|\boldsymbol{F}^{-1}\|^2$ behaves as

$$\|\mathbb{L}(\boldsymbol{x})\| \sim \begin{cases} \mathcal{O}\big(\lambda_{\min}^{-2} \log \frac{1}{\lambda_{\min}}\big), & f = f_{\log}, \\ \mathcal{O}\big(\lambda_{\min}^{-1}\big), & f = f_{\text{quad}}. \end{cases} \tag{37}$$

Thus the logarithmic penalty—benign under tension—produces here the *stronger* blow-up, $\lambda_{\min}^{-2}$ *amplified* by a logarithm, whereas the quadratic penalty gives the milder rate $\lambda_{\min}^{-1}$. This reversal is not a redemption of the quadratic model: $f_{\text{quad}}(J) \to \frac{\kappa}{2} < \infty$ as $J \to 0^+$, so it provides *no energy barrier against volume collapse* and violates Assumption 1; the milder tangent growth merely reflects this missing barrier (the material is free to interpenetrate at finite energy), which is a worse pathology, not a better one. The logarithmic penalty's stronger blow-up is, conversely, the analytic price of a genuine barrier $f_{\log}(J) \to +\infty$—and it is precisely that barrier which forces, and gives physical meaning to, the bifurcation rescue of Section 4.

Ellipticity is *not* lost — the instability is structural. It is tempting to attribute the breakdown to a loss of strong ellipticity (an indefinite acoustic tensor). For this neo-Hookean model that is *false*, and it is worth stating the fact precisely, as it shapes the rest of the analysis.

Lemma 5 (Compressive strong ellipticity of the neo-Hookean tangent)**.** *For the energy* (7)*, the rank-one form of the tangent is exactly* (18)*,* $L_{iJkL}m_iN_Jm_kN_L = \mu_e|\boldsymbol{m}|^2|\boldsymbol{N}|^2 + f''(J)J^2(\boldsymbol{m} \cdot \boldsymbol{F}^{-T}\boldsymbol{N})^2$*. Wherever* $f''(J) \ge 0$ *— in particular throughout compression,* $J \le 1$*, for both penalties* (8) *— the tangent is strongly elliptic with constant* $\ge \mu_e > 0$*, uniformly in* $\boldsymbol{F}$*, and the acoustic tensor* $Q_{ik}(\boldsymbol{N}) = L_{iJkL}N_JN_L$ *is positive definite and never degenerates under compression.*

Proof. The identity is (18) (Lemma 1). As $J \to 0^+$, $f''(J)J^2 \to +\infty$ for $f_{\log}$ (cf. (57)) and $f''(J)J^2 = \kappa J^2 \ge 0$ for f_{quad}; in both cases the second term only *adds* to the constant μ_e. □ □

Thus the compressive corner is *not* a material instability: the volumetric stiffening that blows up $\|\mathbb{L}\|$ is exactly the term $f''(J)J^2$ in (18), and it is positive — it strengthens ellipticity while destroying conditioning. What fails is the *boundedness* of the operator, not its positivity.

Conclusion for Scenario B. The assumption that the deformation smoothly follows the fundamental path into the corner therefore produces, not a loss of ellipticity, but two distinct effects: (i) a blow-up of the tangent *norm*, $\|\mathbb{L}\| \to \infty$ (eq. (37)), which renders the Newton step catastrophically ill-conditioned even though it remains elliptic; and (ii) for a barrier penalty $f_{\log}$, an unphysical infinite accumulation of energy ($f(J) \to \infty$ as $J \to 0$), signalling that the smooth fundamental path is not the physical solution. The genuine mechanism that intervenes is a *structural* bifurcation of the compressed state — the corner analogue of *creasing*, an instability of a material that remains everywhere strongly elliptic (Lemma 5), driven by the geometry and boundary conditions rather than by a change of type of the PDE. To capture it we introduce in Section 4 an imperfect perturbation that regularises the bifurcation and bounds the singularity; we work from here on with the barrier model $f_{\log}$, for which collapse is genuinely resisted.

4 Imperfect bifurcation analysis and asymptotic bounds

The previous section showed that the perfect compressive path drives $\lambda_{\min} \to 0$ and blows up the tangent norm while the material remains strongly elliptic (Lemma 5), so the breakdown is a structural instability rather than a loss of type. The premise of this section is that this scenario is an idealisation: a real system always carries small disturbances—geometric imperfections, load eccentricities, material inhomogeneity—and these select a nearby *regular* solution branch on which the collapse never quite happens. We model the disturbance by a single scalar amplitude ϵ, carry out an imperfect-bifurcation analysis, and quantify how the tangent blows up as $\epsilon \to 0$. The punchline is that the blow-up is confined to a vanishingly small neighbourhood of the corner and proceeds at a controlled algebraic rate, so that for every $\epsilon > 0$ the problem is regular.

4.1 The regularised problem and the fundamental path

We localise to the corner neighbourhood B_ρ of §3, regard the loading as transmitted through data on the outer arc $\partial B_\rho \setminus (\Gamma_D \cup \Gamma_N)$, and introduce a pseudo-time load parameter $t \in [0, 1]$ that monotonically scales the compressive part of that data. Following the answer to the question posed in Scenario B, we add a small, fixed volumetric body force $\epsilon\,\boldsymbol{p}(\boldsymbol{x})$ with $\|\boldsymbol{p}\|_{L^2(B_\rho)} = 1$ and $\epsilon \ll 1$ as the imperfection (we write the field as $\boldsymbol{p}$ to avoid collision with the penalty $f(J)$). The regularised equilibrium is $G(\boldsymbol{u}, t, \epsilon) = 0$,

$$\langle G(\boldsymbol{u}, t, \epsilon), \boldsymbol{v}\rangle = \int_{B_\rho} \boldsymbol{P}(\boldsymbol{F}) : \nabla \boldsymbol{v}\, dV - \epsilon \int_{B_\rho} \boldsymbol{p} \cdot \boldsymbol{v}\, dV - \ell_t(\boldsymbol{v}) = 0 \qquad \forall \boldsymbol{v} \in V, \tag{38}$$

with $\boldsymbol{P} = \partial\Psi/\partial\boldsymbol{F}$, V the admissible test space, and ℓ_t the (linear) load functional at pseudo-time t. Let $\boldsymbol{u}_0(t)$ denote the *fundamental path*, i.e. the branch of (38) with $\epsilon = 0$ that continues the low-load compressive solution. We use the barrier penalty $f_{\log}$ throughout, so the collapse $J \to 0$ is genuinely resisted (Assumption 1); along $\boldsymbol{u}_0(t)$ the corner stretch $\lambda_c(t) := \lambda_{\min}(\boldsymbol{F}_0(t))$ decreases as $t \uparrow 1$ but, on the bifurcating branch we construct, never reaches zero.

Remark 6 (The instability is structural, not material). By Lemma 5 the tangent stays strongly elliptic throughout compression, so the bifurcation here is a *structural* one (a buckling of the boundary-value problem), not a material change of type. We work with the global form on the bounded domain B_ρ. The operator $\mathcal{L}(t) = \partial_{\boldsymbol{u}} G(\boldsymbol{u}_0(t), t, 0)$ has discrete spectrum and attains $\mu_1(t)$ as a genuine eigenvalue *provided* the fundamental stretch stays bounded away from collapse up to the crossing, $\inf_{B_\rho} \lambda_{\min}(\boldsymbol{F}_0(t)) \geq c > 0$ for $t \leq t_c$: then $\mathcal{L}(t)$ has *bounded* strongly-elliptic coefficients on the bounded domain, hence (Gårding, Korn and the Rellich embedding) compact resolvent —

we make this precise as Lemma 14 in the concrete setting. The reservation is essential: what makes the problem delicate is not a loss of ellipticity but the *norm* blow-up $\|\mathbb{L}\| \to \infty$ (eq. (37)), and it is exactly this blow-up, through the scale-invariant (creasing) corner, that would push the bottom of the spectrum into the *essential* spectrum and destroy attainment. Crucially, that blow-up occurs only on the unphysical continuation past t_c, where $\lambda_{\min} \to 0$; at the crossing itself the spectrum is discrete *so long as buckling precedes collapse.* The coercivity constant thus stays positive while the boundedness constant diverges, and the crossing eigenvalue $\mu_1(t)$ and the conditioning are controlled by different mechanisms. Assumption 6 records the spectral facts (attainment and simplicity) we use; §5.10 shows that, for the concrete model, both follow once buckling is known to precede collapse.

4.2 What is provable, and what must be assumed

It is worth separating the genuinely provable facts about the critical point from the irreducible hypotheses. Let $Q_t(\boldsymbol{v}) := \int_{B_\rho} \nabla \boldsymbol{v} : \mathbb{L}(\boldsymbol{F}_0(t)) : \nabla \boldsymbol{v} \, dV$ be the second-variation form on the fundamental path and $\mu_1(t) := \inf_{\|\boldsymbol{v}\|_V = 1} Q_t(\boldsymbol{v})$.

The critical load exists (conditionally provable). A clarification is essential here. By Lemma 5 the neo-Hookean material remains strongly elliptic throughout compression, so $\mu_1(t)$ *cannot* be driven negative by a rank-one (material) instability: any crossing must be a *structural* loss of positivity of the global form Q_t, i.e. a buckling of the boundary-value problem in which the destabilising test field is not rank-one. This is the creasing mechanism.

Lemma 6 (Existence of the critical load). *Assume (i) coercivity at zero load, $\mu_1(0) > 0$, and (ii) a structural instability of the compressed state: there exist $t_* \in (0, 1)$ and $\boldsymbol{v}_* \in V$ with $Q_{t_*}(\boldsymbol{v}_*) < 0$. Then μ_1 is continuous on $[0, t_*]$ and there is a first $t_c \in (0, t_*]$ with $\mu_1(t_c) = 0$ and $\mu_1(t) > 0$ for $t < t_c$.*

Proof. At $t = 0$, $\boldsymbol{F}_0 = \boldsymbol{I}$ and $\mathbb{L} = \mu_e \mathbb{I} +$ (volumetric at J=1) is positive definite, so with Korn's inequality $\mu_1(0) > 0$. Hypothesis (ii) gives $\mu_1(t_*) \le Q_{t_*}(\boldsymbol{v}_*)/\|\boldsymbol{v}_*\|_V^2 < 0$. As $t \mapsto \mathbb{L}(\boldsymbol{F}_0(t))$ is continuous, so is μ_1, and the intermediate value theorem gives the first zero t_c. □ □

Hypothesis (i) is provable for neo-Hookean (positive-definiteness at $\boldsymbol{F} = \boldsymbol{I}$ plus Korn). Hypothesis (ii) — the existence of a structurally destabilising field — is the genuine content and is *not* elementary: it is the corner counterpart of the Biot/creasing instability [1, 3] and we do not establish it here.

The critical mode (assumed).

Assumption 6 (Critical mode)**.** At t_c the infimum $\mu_1(t_c) = 0$ is attained at a mode $\boldsymbol{\phi} \in V$, $\|\boldsymbol{\phi}\|_V = 1$, spanning a one-dimensional, isolated kernel of $\mathcal{L}_c$ (the rest of the spectrum bounded away from 0), with $\boldsymbol{\phi}$ supported in the edge neighbourhood B_ρ.

This bundles two spectral facts: *attainment* (the minimising sequence does not escape into the corner as essential spectrum) and *simplicity.* Simplicity is generic and, for the model bodies, can be argued from the lateral *face-preserving* reflection R of §5.6 (*not* the corner bisector $\theta \mapsto \frac{\pi}{2} - \theta$, which swaps Γ_D and Γ_N and is no symmetry of the mixed problem); attainment is the borderline-elliptic difficulty flagged in Remark 6. Neither is irreducible: in the concrete setting attainment is *proved* (Lemma 14) and simplicity is reduced to genericity *within* the symmetry sector carrying the critical mode (§5.6), the only residue being the cleaner geometric condition that buckling precede pointwise collapse.

The crossing is transversal (provable under monotone loading).

Lemma 7 (Transversality)**.** *Suppose the loading is monotonically destabilising at t_c,*

$$\langle \boldsymbol{\phi}, \dot{\mathbb{L}}(t_c)\boldsymbol{\phi}\rangle := \int_{B_\rho} \nabla\boldsymbol{\phi} : \partial_t \mathbb{L}(\boldsymbol{F}_0(t_c)) : \nabla\boldsymbol{\phi}\, dV < 0. \tag{M}$$

Then, under Assumption 6, μ_1 is differentiable at t_c with $a := -\mu_1'(t_c) = -\langle \boldsymbol{\phi}, \dot{\mathbb{L}}(t_c)\boldsymbol{\phi}\rangle > 0$.

Proof. For a simple eigenvalue, first-order perturbation theory gives $\mu_1'(t_c) = \langle \boldsymbol{\phi}, \dot{\mathbb{L}}(t_c)\boldsymbol{\phi}\rangle$; hypothesis (M) fixes the sign. □ □

Condition (M) is the precise meaning of "more compression $\Rightarrow$ less stiffness in the buckling direction" and holds for any monotone compressive load path; it is an input on the *loading*, not on the solution.

The imperfection is active (provable, generic).

Lemma 8 (Generic activation)**.** *For every $\boldsymbol{p}$ outside the closed codimension-one subspace $\boldsymbol{\phi}^\perp$, the projection $\eta := \int_{B_\rho} \boldsymbol{p}\cdot\boldsymbol{\phi}\, dV \neq 0$. Hence $\eta \neq 0$ for generic imperfection.*

Remark 7 (Universality of the imperfection)*.* The volumetric body force $\epsilon\,\boldsymbol{p}$ is chosen only for concreteness; the analysis depends on the imperfection *solely* through the scalar $\eta = \langle \boldsymbol{p}, \boldsymbol{\phi}\rangle$, so any perturbation with a nonzero $\boldsymbol{\phi}$-projection produces the identical reduced equation (41). Indeed, write a generic imperfect equilibrium as $G(\boldsymbol{u}, t, 0) = \epsilon\,\boldsymbol{h}$ with $\boldsymbol{h} \in V'$ the (linearised) imperfection functional. A boundary-traction defect on Γ_N contributes $\boldsymbol{h}(\boldsymbol{v}) = \int_{\Gamma_N} \boldsymbol{t}_1 \cdot \boldsymbol{v}\, dS$; a geometric imperfection — rounding the corner with radius $\sim \epsilon$, or a small shape variation $\boldsymbol{x} \mapsto \boldsymbol{x} + \epsilon\,\boldsymbol{\theta}(\boldsymbol{x})$ — enters, after pulling back to the reference domain, as a fixed load functional $\boldsymbol{h}(\boldsymbol{v}) = \int_{B_\rho} \mathbb{S}[\boldsymbol{\theta}] : \nabla\boldsymbol{v}\, dV$ linear in the shape field $\boldsymbol{\theta}$. In every case the kernel projection $\langle G, \boldsymbol{\phi}\rangle = 0$ yields the same normal form with η replaced by $\langle \boldsymbol{h}, \boldsymbol{\phi}\rangle$, nonzero for $\boldsymbol{h}$ outside the closed codimension-one subspace $\boldsymbol{\phi}^\perp$ (Lemma 8). The Koiter $\frac{2}{3}$ law (44) and all subsequent bounds are therefore universal across generic imperfections, not an artefact of the body-force model.

The nonlinearity is explicit; only the sign of its orthogonal-mode feedback is open. For the neo-Hookean model the reduced coefficients are not opaque. By Lemma 13 the deviatoric energy is quadratic and drops out, so every coefficient of order ≥ 3 is purely volumetric; moreover the quadratic coefficient $c_2 := \frac{1}{2}\langle \boldsymbol{\phi}, \partial_{\boldsymbol{u}}^2 G[\boldsymbol{\phi}, \boldsymbol{\phi}]\rangle$ vanishes *identically* for a mode antisymmetric under the lateral, face-preserving reflection R of §5.6 — not the corner bisector $\theta \mapsto \frac{\pi}{2} - \theta$, which swaps Γ_D and Γ_N and is no symmetry at all. What these structural facts do *not* settle is the *sign* of the cubic and quintic coefficients; indeed, by Lemma 13(ii), the explicit single-mode (frozen) parts point *against* subcriticality, $b_{\mathrm{dir}} > 0$ and $d_{\mathrm{dir}} < 0$ in compression. The subcritical pattern therefore hinges on the geometry-dependent orthogonal-mode feedback overturning these signs; we record it as a residual hypothesis to be verified numerically (Remark 8) rather than asserted from the creasing analogy [1, 3].

Assumption 7 (Subcritical sign pattern (residual, geometry-dependent))**.** With $c_2 = 0$ by Lemma 13(i), the full reduced cubic and quintic coefficients

$$b := \tfrac{1}{6}\langle \boldsymbol{\phi}, \partial_{\boldsymbol{u}}^3 G[\boldsymbol{\phi}, \boldsymbol{\phi}, \boldsymbol{\phi}]\rangle, \qquad d := \tfrac{1}{120}\langle \boldsymbol{\phi}, \partial_{\boldsymbol{u}}^5 G[\boldsymbol{\phi}, \dots, \boldsymbol{\phi}]\rangle, \tag{39}$$

each the frozen part of Lemma 13 corrected by the orthogonal-mode feedback, satisfy $b < 0 < d$ (subcritical with quintic saturation). This is a condition on the corner geometry, not on the material model.

Remark 8 (Status of the subcritical signs: geometry-decided, checked numerically). Assumption 7 is the one genuinely residual hypothesis of the construction, and Lemma 13 localises exactly where its content lies. The deviatoric response and the quadratic coefficient are settled ($c_2 = 0$, no feedback correction); the cubic and quintic decompose as $b = b_{\mathrm{dir}} + b_\psi$ and $d = d_{\mathrm{dir}} + d_\psi$, with the single-mode parts explicit and, in compression, of definite but *adverse* sign ($b_{\mathrm{dir}} = 2\int_{B_\rho} f''(J_0) d_\phi^2 \, dV > 0$, $d_{\mathrm{dir}} = \int_{B_\rho} f'''(J_0) d_\phi^3 \, dV < 0$). The corrections b_ψ, d_ψ are the Lyapunov–Schmidt feedback through the orthogonal complement $V^\perp$: quadratic forms in the second-order field $\boldsymbol{\psi}$ that depend on the opening angle, the Dirichlet–Neumann arrangement and the mode $\boldsymbol{\phi}$ — on the geometry, not on f. Subcriticality thus requires b_ψ to overturn the explicit positive b_{dir} (and likewise for d). The reason this does not reduce to a single universal corner number is a homogeneity count (Proposition 6, §5.7): the clamped–free edge exponent satisfies $\alpha > \frac{1}{2}$, so the cubic integral is convergent at the tip but *not* dominated by it, leaving an $\mathcal{O}(1)$ dependence on the normalised stress-intensity $\hat{K}$; the sign of b is therefore a function $b(\nu, \hat{K})$ on the model wedge rather than an inheritance from flat creasing. We do not sign b_ψ, d_ψ analytically; they are evaluated numerically from the eigenmode and reduced operator of the two-dimensional model wedge of Proposition 5 as a function of $(\nu, \hat{K})$, which is the proper arbiter of a geometry-dependent sign. Were the verdict $b > 0$ instead, the bifurcation would be supercritical and *more* benign — the physical branch smooth through t_c with no snap at onset — and the regularisation conclusions of §5.10 and Proposition 4 would hold a fortiori; the complementary transcritical case $c_2 \neq 0$ is covered in Remark 9.

4.3 Lyapunov–Schmidt reduction

Proposition 1 (Normal form)**.** *Under Assumption 6, Lemma 7 and Assumption 7, there is a neighbourhood of* $(\boldsymbol{u}_0(t_c), t_c, 0)$ *in which the solutions of* (38) *are exactly*

$$\boldsymbol{u}(t,\epsilon) = \boldsymbol{u}_0(t) + w\,\boldsymbol{\phi} + \boldsymbol{\psi}(w,t,\epsilon), \qquad \boldsymbol{\psi} \in V^\perp := \{\boldsymbol{\phi}\}^\perp, \tag{40}$$

where $\boldsymbol{\psi}$ *is smooth with* $\boldsymbol{\psi} = \mathcal{O}(w^2 + \epsilon)$*, and the amplitude* w *solves the scalar imperfect bifurcation equation, retained to quintic order because the cubic is destabilising* ($b < 0$)*,*

$$a\,(t_c - t)\,w + b\,w^3 + d\,w^5 = \epsilon\,\eta + \mathcal{O}\big(\epsilon^2 + \epsilon w + |t_c - t|\,w^2 + w^6\big), \tag{41}$$

with $a > 0$ *(Lemma 7),* $b < 0 < d$ *(Assumption 7) and* η *as in Lemma 8.*

Proof. Let P be the V-orthogonal projection onto $\operatorname{span}\{\boldsymbol{\phi}\}$ and $Q = I - P$. Write $\boldsymbol{u} = \boldsymbol{u}_0(t) + w\boldsymbol{\phi} + \boldsymbol{\psi}$ with $\boldsymbol{\psi} \in V^\perp$. The range equation $Q\,G = 0$ has invertible linearisation $Q\,\mathcal{L}_c\,Q$ on $V^\perp$ by Assumption 6, so the implicit function theorem yields a unique smooth $\boldsymbol{\psi} = \boldsymbol{\psi}(w,t,\epsilon)$ with $\boldsymbol{\psi}(0,t_c,0) = 0$ and $\boldsymbol{\psi} = \mathcal{O}(w^2 + \epsilon + |t_c - t|)$. Substituting into the kernel equation $\langle G, \boldsymbol{\phi}\rangle = 0$ and Taylor-expanding G about $(\boldsymbol{u}_0(t_c), t_c, 0)$ gives (41): the linear term $a(t_c - t)w$ from Lemma 7, the cubic bw^3 and quintic dw^5 from Assumption 7 (the quadratic vanishing by symmetry), and the inhomogeneity $\epsilon\eta$ from $-\epsilon\langle \boldsymbol{p}, \boldsymbol{\phi}\rangle$. Because $b < 0$, the cubic alone does not saturate the amplitude, which is why the quintic dw^5 ($d > 0$) is retained. □ □

4.4 Subcritical branch structure and imperfection sensitivity

Perfect system. Set $\epsilon = 0$ and $t < t_c$. Dividing (41) by w gives $a(t_c - t) + bw^2 + dw^4 = 0$, i.e.

$$w^2 = \frac{-b \pm \sqrt{b^2 - 4ad(t_c - t)}}{2d}, \tag{42}$$

which (with $a > 0$, $b < 0$, $d > 0$) admits real nontrivial branches only for $t \geq t_L$, where

$$t_L := t_c - \frac{b^2}{4ad} < t_c, \qquad w_L^2 = \frac{-b}{2d} = \mathcal{O}(1). \tag{43}$$

At t_L the two branches meet in a limit point (fold) at the *finite* amplitude w_L. The fundamental branch $w = 0$ is stable for $t < t_c$ and unstable for $t > t_c$; the upper nontrivial branch is stable. Hence on $[t_L, t_c)$ the compressed state and a finite-amplitude creased state coexist: the bifurcation is subcritical, with hysteresis and a *snap* at t_c.

Imperfect system.

Proposition 2 (Imperfection sensitivity, Koiter $\frac{2}{3}$ law)**.** *For $\epsilon > 0$ the compressed branch terminates at a limit point that issues from the bifurcation point $(0, t_c)$ and is depressed below t_c according to*

$$t_s(\epsilon) = t_c - C\,|\eta\,\epsilon|^{2/3} + o(\epsilon^{2/3}), \qquad C = \frac{3\,|b|^{1/3}}{2^{2/3}\,a} > 0, \tag{44}$$

so the load at which the compressed branch ceases to exist is reduced by $\mathcal{O}(\epsilon^{2/3})$. The turning amplitude there is itself small, $w_* = \mathcal{O}(\epsilon^{1/3})$; the $\mathcal{O}(1)$ amplitude is that of the creased branch onto which the structure then snaps.

Proof. Write (41) as $g(w, t) = 0$ with

$$g(w, t) := a(t_c - t)\,w + b\,w^3 + d\,w^5 - \epsilon\eta,$$

the analytic remainder of (41) being subdominant at the scaling found below (we verify this at the end). A limit point in the load is a point of the solution set at which the branch turns, i.e. at which t is stationary along the curve; differentiating $g = 0$ implicitly, this is the pair

$$g(w, t) = 0, \qquad g_w(w, t) = a(t_c - t) + 3b\,w^2 + 5d\,w^4 = 0. \tag{45}$$

We follow the limit point that issues from the bifurcation point $(0, t_c)$ as $\epsilon \to 0$; this is the one that caps the slightly deflected compressed branch and hence sets the snap load.

Turning amplitude. Using the second equation of (45) to eliminate the load, $a(t_c - t) = -(3b\,w^2 + 5d\,w^4)$, and substituting into the first gives a single equation for the turning amplitude,

$$-2b\,w^3 - 4d\,w^5 = \epsilon\eta. \tag{46}$$

Because $b < 0$, the small root of (46) as $\epsilon \to 0$ obeys $-2b\,w^3 = \epsilon\eta\,(1 + o(1))$, the quintic entering only at relative order w^2; hence

$$w_*(\epsilon) = \left(\tfrac{|\eta|\,\epsilon}{2|b|}\right)^{1/3}(1 + o(1)) = \mathcal{O}(\epsilon^{1/3}), \tag{47}$$

in agreement with the unstable-branch scaling (48) evaluated at the turn. In particular $w_* \to 0$, so at this limit point the cubic dominates the quintic, $d\,w_*^4/(b\,w_*^2) = \mathcal{O}(\epsilon^{2/3})$.

Snap load. Inserting (47) into $g_w = 0$ and discarding the quintic, a relative $\mathcal{O}(\epsilon^{2/3})$ correction,

$$a\big(t_c - t_s(\epsilon)\big) = -3b\,w_*^2\,(1 + o(1)) = 3|b|\left(\tfrac{|\eta|\,\epsilon}{2|b|}\right)^{2/3}(1 + o(1)),$$

which is (44) with $C = 3|b|^{1/3}/(2^{2/3}a) > 0$, the classical Koiter $\frac{2}{3}$ power law [5, 2].

Remainder. At the limit point $w = \mathcal{O}(\epsilon^{1/3})$ and $t_c - t = \mathcal{O}(\epsilon^{2/3})$, so the omitted terms $\mathcal{O}(\epsilon^2 + \epsilon w + |t_c - t|\,w^2 + w^6)$ of (41) are $\mathcal{O}(\epsilon^{4/3})$, a relative $\mathcal{O}(\epsilon^{1/3})$ against the dominant balance $a(t_c - t)w \sim b\,w^3 \sim \epsilon\eta = \mathcal{O}(\epsilon)$. They perturb w_* and $t_s(\epsilon)$ only within the stated $o(\epsilon^{2/3})$, leaving the exponent and C unchanged. □ □

The unstable continuation branch. Separately, near t_c the small-amplitude (backward-bending, unstable) branch of (41) satisfies, at $t = t_c$, $b\,w^3 = \epsilon\eta\,(1 + o(1))$ with the quintic subdominant, so

$$|w(\epsilon)| = \Big(\frac{|\eta|}{|b|}\Big)^{1/3} \epsilon^{1/3}\,(1 + o(1)), \qquad \epsilon \to 0. \tag{48}$$

This is the branch a displacement- or arc-length-controlled solver tracks through the instability, and it is the one that feeds the tangent estimate below; it is *not* the branch a load-controlled physical system follows.

Remark 9 (Robustness to the bifurcation type: pitchfork vs. transcritical). Assumption 7 takes the reduced potential to be odd ($c_2 = 0$), a subcritical *pitchfork*; by the discussion there this holds when the critical mode is antisymmetric under the lateral reflection R of §5.6. If that symmetry is absent or the ground mode lies in the symmetric sector, the quadratic coefficient $c_2 \neq 0$ survives and (41) is instead *transcritical*,

$$a\,(t_c - t)\,w + c_2\,w^2 + \mathcal{O}(w^3) = \epsilon\,\eta. \tag{49}$$

The qualitative architecture of the section is unaffected. Repeating the limit-point analysis of Proposition 2 on (49) (solve $g = g_w = 0$ with $g = a(t_c - t)w + c_2 w^2 - \epsilon\eta$) gives a turning amplitude $w_*^2 = -\epsilon\eta/c_2 = \mathcal{O}(\epsilon)$ and a snap load depressed by the classical Koiter *one-half* law,

$$t_s(\epsilon) = t_c - C'\,|\eta\,\epsilon|^{1/2} + o(\epsilon^{1/2}), \tag{50}$$

in place of the $\frac{2}{3}$ law and $w_* = \mathcal{O}(\epsilon^{1/3})$ of the pitchfork. The determinant relief (52)–(54) still lifts $\lambda_{\min}$ off zero, now at order $w_*^2 = \mathcal{O}(\epsilon)$ rather than $\mathcal{O}(\epsilon^{2/3})$, so the regularised tangent bound of §5.10 reads $\|\mathbb{L}\| = \mathcal{O}(\epsilon^{-2}\log\frac{1}{\epsilon})$ in place of $\mathcal{O}(\epsilon^{-4/3})$, and the localisation of Proposition 4 persists with the correspondingly shifted exponents. In particular the central result — a bounded physical branch carrying a vanishingly localised, regularised tangent blow-up for every $\epsilon > 0$ — is insensitive to whether the corner bifurcation is a subcritical pitchfork or a transcritical fold; only the imperfection-sensitivity exponents change.

4.5 Kinematic relief: lower bound on J

The heart of the regularisation is geometric: the buckling mode trades pure compression for rotation and shear, which lifts the determinant off zero. The reason the relief is second order in w—and hence why the scaling exponents below are what they are—is that the mode is, to leading order, a *rotation*, and rotations do not change volume at first order. We do *not* assume the first-order volume change vanishes exactly; we use the quantitative bound on it posited by Assumption 8 (the asymptotic-rotation hypothesis, verified numerically in Remark 10) and show it is dominated by the second-order relief.

Lemma 9 (Determinant relief)**.** *Let $\boldsymbol{F} = \boldsymbol{F}_0 + w\,\nabla\boldsymbol{\phi} + \mathcal{O}(w^2)$ at the corner, with $\boldsymbol{F}_0 = \boldsymbol{F}_0(t)$ on the loaded path, $J_0 = \det \boldsymbol{F}_0$, and write the first-order volume change as*

$$D_1 := \operatorname{cof} \boldsymbol{F}_0 : \nabla\boldsymbol{\phi} = J_0\,\boldsymbol{F}_0^{-T} : \nabla\boldsymbol{\phi}. \tag{51}$$

By the asymptotic-rotation estimate (58) of Assumption 8, the critical mode is asymptotically volume-preserving*, $|D_1| \le c_\phi\,J_0/\sqrt{|\log J_0|}$, so D_1 does not vanish at finite load but is itself small with J_0. Let $d_\phi := \frac{1}{2}\,\frac{d^2}{dw^2}\big|_0 \det(\boldsymbol{F}_0 + w\,\nabla\boldsymbol{\phi})$ be the second-order coefficient of the determinant — the second mixed discriminant of $\boldsymbol{F}_0$ ($d-2$ copies) with $\nabla\boldsymbol{\phi}$ (two copies), which for $d = 2$ reduces to*

$\det(\nabla\boldsymbol{\phi})$. *We take the mode to satisfy the relief-orienting condition* $d_\phi > 0$ *a.e.* on a neighbourhood of the collapse locus Σ_{c} *— the only region where the relief floor is invoked (*(72) *in §5.8; a property of the mode,* not *supplied by the transversality Lemma 11). Away from* Σ_{c}*, where* J_0 *stays bounded below, the sign of* d_ϕ *is immaterial: the finite-*J_0 *case (i) below absorbs a negative* $w^2 d_\phi$ *into the* J_0 *margin. Then*

$$J = \det \boldsymbol{F} = J_0 + w\, D_1 + w^2\, d_\phi + \mathcal{O}(w^3), \tag{52}$$

and the linear term is dominated by the other two: for $|w| \le w_0$ *small,*

$$J \;\ge\; \tfrac{1}{2}\, J_0 + \tfrac{1}{2}\, w^2\, d_\phi. \tag{53}$$

Consequently, with $\lambda_c = \lambda_{\min}(\boldsymbol{F}_0)$ *and the remaining* $d-1$ *stretches of* $\boldsymbol{F}$ *bounded,* $\prod_{i\ge 2} \lambda_i(\boldsymbol{F}) = \mathcal{O}(1)$*,*

$$\lambda_{\min}(\boldsymbol{F}) = \frac{J}{\prod_{i\ge 2}\lambda_i(\boldsymbol{F})} \;\gtrsim\; C_1\lambda_c + C_2\, w^2, \qquad C_1, C_2 > 0. \tag{54}$$

Proof. Expanding the determinant in w, $\det(\boldsymbol{F}_0 + w\,\nabla\boldsymbol{\phi}) = \sum_{k=0}^{d} w^k\, D_k(\boldsymbol{F}_0, \nabla\boldsymbol{\phi})$ with D_k the k-th mixed discriminant; $D_0 = \det \boldsymbol{F}_0 = J_0$, $D_1 = \operatorname{cof} \boldsymbol{F}_0 : \nabla\boldsymbol{\phi}$, and $D_2 = d_\phi$, giving (52) (the $\mathcal{O}(w^3)$ collecting $D_3, \ldots, D_d$). For (53) we bound the linear term wD_1 in two complementary regimes; in both, $\mathcal{O}(w^3)$ is absorbed into the constants for $|w| \le w_0$.

(i) Finite J_0 *(at and near the bifurcation, and everywhere off* Σ_{c}*).* Here $J_0 \ge c > 0$ and D_1 is finite, so the bound needs no smallness of D_1, only of w: choosing $w_0 \le J_0/(4|D_1|)$ gives $|w\, D_1| \le \frac{1}{4} J_0$ for $|w| \le w_0$, whence $J = J_0 + wD_1 + w^2 d_\phi + \mathcal{O}(w^3) \ge \frac{3}{4} J_0 + \frac{1}{2} w^2 d_\phi$. *No sign of* d_ϕ *is used here*: if $d_\phi \ge 0$ this is already $\ge \frac{1}{2} J_0 + \frac{1}{2} w^2 d_\phi$, the relief bound (53); if $d_\phi < 0$ (the collapse-driving orientation, which the computation finds confined to a thin surface layer, §5.8) shrink w_0 further to $w_0 \le \sqrt{J_0/(4|d_\phi|)}$ so that $\frac{1}{2} w^2 |d_\phi| \le \frac{1}{4} J_0$, whence $J \ge \frac{3}{4} J_0 - \frac{1}{4} J_0 = \frac{1}{2} J_0$ and (54) holds with the load floor $C_1\lambda_c$ alone (no relief, but no collapse either). This is the regularity-only argument: at t_c, where $J_0(t_c) > 0$, and at every point off the collapse locus, (53) (resp. its $d_\phi < 0$ surrogate $J \ge \frac{1}{2} J_0$) holds with no rotation condition and with no sign hypothesis on d_ϕ (cf. Lemma 10).

(ii) Near collapse ($J_0 \to 0$*, on the tube around* Σ_{c}*).* Now w_0 from (i) would shrink with J_0, so we use instead the asymptotic-rotation bound $|D_1| \le c_\phi J_0/\sqrt{|\log J_0|}$ of Assumption 8, *together with* the localised orientation $d_\phi > 0$ of (72) — the one region where that sign is genuinely invoked, as the denominator in Young's inequality $|w\, D_1| \le \frac{1}{2} d_\phi w^2 + \frac{1}{2d_\phi} D_1^2$. With $D_1^2 \le c_\phi^2 J_0^2/|\log J_0| \le d_\phi J_0$ once J_0 is small (the factor $J_0/|\log J_0| \to 0$), $\frac{1}{2d_\phi} D_1^2 \le \frac{1}{2} J_0$ and $|w\, D_1| \le \frac{1}{2} d_\phi w^2 + \frac{1}{2} J_0$, and inserting into (52) yields (53). Thus the residual first-order term, far from dominating, is swallowed by the barrier-controlled J_0 and the relief $w^2 d_\phi$ — the precise content of "buckling relieves compression," with no appeal to an exact rotation condition. Since $J = \prod_i \lambda_i$ and the $d-1$ largest stretches are $\mathcal{O}(1)$ on the relieved branch (the buckling redirects, but does not unboundedly stretch, the material near the corner), (52) gives (54). □ □

The estimate (54) carries two competing floors, $C_1\lambda_c$ and $C_2 w^2$, and which one binds depends on the load. At the bifurcation point itself the perfect stretch $\lambda_c(t_c) > 0$ is finite (buckling precedes collapse, Remark 6); there $C_1\lambda_c$ dominates, $\lambda_{\min}$ is bounded below by a constant, and *no* blow-up occurs — consistently with the bounded-tangent physical branch of Proposition 3. The tangent divergence is felt only along the *unstable continuation* that a solver tracks to deeper loads, where the perfect path drives $\lambda_c(t) \to 0$; precisely in the regime $\lambda_c \lesssim w^2$ the relief term takes over. Combining (54) with the branch amplitude $w \sim \epsilon^{1/3}$ of (48) then gives, in that near-collapse regime,

$$\lambda_{\min}(\epsilon) \gtrsim C_2\, w(\epsilon)^2 \sim \mathcal{O}(\epsilon^{2/3}), \qquad \|\boldsymbol{F}^{-1}\| = \lambda_{\min}^{-1} \sim \mathcal{O}(\epsilon^{-2/3}). \tag{55}$$

Outside that regime ($\lambda_c \gtrsim \epsilon^{2/3}$) the floor $C_1\lambda_c$ keeps $\|\boldsymbol{F}^{-1}\|$ even smaller, so (55) is the worst case. On the stable physical branch the picture is milder still; we contrast the two after Theorem 2.

We now examine the smallness of the first-order volume change D_1 in (51). The exact contraction of the tangent (10) gives the second variation of Ψ on the fundamental state $\boldsymbol{F}_0$ in a direction $\boldsymbol{\phi}$ (plane strain),

$$\langle \boldsymbol{\phi}, \mathcal{L}\boldsymbol{\phi}\rangle = \mu_e\,|\nabla\boldsymbol{\phi}|^2 + f''(J_0)\,(\delta J)^2 + 2\,f'(J_0)\,\det(\nabla\boldsymbol{\phi}), \qquad \delta J = \operatorname{cof}\boldsymbol{F}_0 : \nabla\boldsymbol{\phi}. \tag{56}$$

For $f_{\log}$ both volumetric prefactors diverge as $J_0 \to 0$,

$$f''(J_0) = \frac{\mu_e + 2\lambda_e - 2\lambda_e \log J_0}{J_0^2} \sim \frac{2\lambda_e|\log J_0|}{J_0^2}, \qquad f'(J_0) \sim \frac{2\lambda_e \log J_0}{J_0} \to -\infty. \tag{57}$$

Crucially, the determinant term $2f'(J_0)\det(\nabla\boldsymbol{\phi})$, with prefactor $\sim |\log J_0|/J_0$, is *more* singular than the term $f''(J_0)(\delta J)^2 \sim |\log J_0|$ (recall $\delta J \sim J_0 \cdot \mathcal{O}(1)$). Hence δJ cannot be bounded by isolating the f'' term and discarding the rest; the two regimes below treat its smallness honestly.

Lemma 10 (Bifurcation regularity)**.** *At the bifurcation load t_c the fundamental determinant is bounded below, $J_0(t_c) > 0$ (buckling precedes collapse, Remark 6). Then $f'(J_0), f''(J_0)$ are finite,* (56) *is a regular quadratic form, and the neutral mode satisfies $\langle \boldsymbol{\phi}, \mathcal{L}\boldsymbol{\phi}\rangle = 0$ with δJ and* $\det\nabla\boldsymbol{\phi}$ *finite. No rotation condition is needed at t_c: the relief bound* (53) *holds there by the finite-J_0 case (i) of Lemma 9, which uses only $J_0(t_c) > 0$ and smallness of w (not of D_1).*

Along the *unstable continuation* the fundamental state is driven toward collapse, $J_0 \to 0$, and there the relief estimate (53) requires the residual first-order term to stay subordinate. By (56) this is no longer automatic, so we isolate it as a hypothesis on the low-energy modes, verified numerically below in place of the (false) claim that it follows from the f'' term alone.

Assumption 8 (Asymptotic rotation, (AR))**.** Along the near-collapse continuation the buckling mode $\boldsymbol{\phi}$ keeps $\langle \boldsymbol{\phi}, \mathcal{L}\boldsymbol{\phi}\rangle$ bounded through cancellation in (56) (the shear term balancing the singular determinant term), and its first-order volume change obeys

$$\delta J = \operatorname{cof}\boldsymbol{F}_0 : \nabla\boldsymbol{\phi} = \mathcal{O}\Big(\frac{J_0}{\sqrt{|\log J_0|}}\Big) \to 0 \qquad (J_0 \to 0^+), \tag{58}$$

so the mode is asymptotically volume-preserving at the rate that makes the relief term $w^2 d_\phi$ dominate the residual first-order term in (52).

Remark 10 (Numerical verification of (AR))*.* The *qualitative* content of (AR) is supported by the consistent weak solver of §6.3 (`graded_buckling.py`), evaluated on the *same* critical mode that fixes the subcritical sign. The neutral co-energy vanishes, $\langle \boldsymbol{\phi}, \mathcal{L}\boldsymbol{\phi}\rangle \approx 10^{-8}$, confirming Lemma 10; the dimensionless rotation defect $\rho_{\mathrm{rot}} := \|\delta J\|_2/\|\nabla\boldsymbol{\phi}\|_2 \approx 0.08$–$0.16$ across the $(\nu, \hat{K})$ rectangle, and remains small and stable (≈ 0.07–0.09) as the base compression is deepened, so the mode is indeed nearly volume-preserving. The decomposition also makes the mechanism explicit: the determinant term is 6–10× the $f''(\delta J)^2$ term—so the latter cannot be isolated, exactly as (56) warns—and the bounded co-energy is achieved by cancellation against the shear term, the structural content of (AR). We stress what these data do and do not establish: they confirm that the critical mode is nearly volume-preserving with bounded co-energy by cancellation, but they do *not* verify the precise J_0-dependent rate $\delta J = \mathcal{O}(J_0/\sqrt{|\log J_0|})$ of (58) — the accessible base states reach only $J_0 \approx 0.56$, not the asymptotic limit $J_0 \to 0^+$. That rate, which is what Lemma 9 and Theorem 2 actually invoke, remains a genuine assumption, consistent with but not proved by the computation.

4.6 Regularised tangent bound and spatial localisation

Assumption 9 (Non-degeneracy of the stable branch, (ND))**.** The stable creased equilibrium reached after the snap has bounded distortion uniformly in ϵ: there is $c' > 0$ with $\lambda_{\min}(\boldsymbol{F}) \geq c'$ (equivalently $J = \det \boldsymbol{F} \geq J_{\text{floor}} > 0$ pointwise) and the remaining $d-1$ stretches are $\mathcal{O}(1)$, throughout B_ρ.

We state (ND) as a hypothesis, not a consequence of the energy bound: an integral bound $\int f(J) < \infty$ gives only measure-type control of $\{J \text{ small}\}$, not a pointwise floor. It is the finite-amplitude analogue of the small-w relief of Lemma 9, which it does *not* invoke — the lemma is local ($|w| \leq w_0$) while the snap is to $\mathcal{O}(1)$ amplitude. Physically it is the standard expectation for creasing: the post-snap state is a finite-amplitude fold of bounded distortion, not a collapse, and the barrier $f_{\log}(J) \to \infty$ makes collapse energetically prohibitive. We emphasise that, unlike (AR), it is *not* checked by the spectral computation of §6.3: that computation reaches the critical mode and the subcritical sign but does not track the nonlinear branch past the snap. A direct verification would require a full nonlinear path-following continuation onto the $\mathcal{O}(1)$ creased branch, which we do not carry out here; (ND) is therefore posited, on physical grounds, and left as the one regularity hypothesis of the physical-branch statement.

Proposition 3 (Stable physical branch: bounded tangent)**.** *Under load control the structure follows the compressed fundamental branch up to the snap at $t_s(\epsilon) \leq t_c$ (Proposition 2) and then jumps to the creased branch of amplitude $w = \mathcal{O}(1)$. On the compressed branch the bound is unconditional: $\lambda_{\min} \geq \lambda_c(t) \geq \lambda_c(t_c) > 0$ (buckling precedes collapse), so $\|\mathbb{L}\| = \mathcal{O}(1)$ there. Under the non-degeneracy hypothesis 9 the post-snap branch is likewise controlled, and on the whole physical branch*

$$\max_{\boldsymbol{x}\in B_\rho} \|\mathbb{L}(\boldsymbol{x}, \epsilon)\| = \mathcal{O}(1) \quad \textit{uniformly in } \epsilon, \tag{59}$$

so the sole effect of the imperfection is the $\mathcal{O}(\epsilon^{2/3})$ reduction (44) *of the snap load: there is no tangent blow-up on the physical branch.*

Theorem 2 (Unstable continuation branch: regularised blow-up)**.** *Under Assumptions 1, 6, 7, 8 and the conclusions* (55) *of Lemma 9, with the barrier penalty $f_{\log}$, the tangent* along the unstable continuation branch (48)*—the branch tracked by a displacement- or arc-length-controlled solver—satisfies*

$$\max_{\boldsymbol{x}\in B_\rho} \|\mathbb{L}(\boldsymbol{x}, \epsilon)\| \;\leq\; C\,\Theta(J)\,\|\boldsymbol{F}^{-1}\|^2 \;\sim\; \mathcal{O}\big(\epsilon^{-4/3}\log\tfrac{1}{\epsilon}\big), \qquad \epsilon \to 0. \tag{60}$$

Proof. By (11), $\|\mathbb{L}\| \leq \mu_e + \Theta(J)\|\boldsymbol{F}^{-1}\|^2$. For $f_{\log}$, $\Theta(J) = \mathcal{O}(\log\frac{1}{J})$ as $J \to 0$ by (36), and on the unstable branch $J \sim \lambda_{\min} \sim \epsilon^{2/3}$ by (55), so $\Theta(J) = \mathcal{O}(\log\frac{1}{\epsilon})$; multiplying by $\|\boldsymbol{F}^{-1}\|^2 \sim \epsilon^{-4/3}$ gives (60). □ □

The two results together are the precise sense in which "the singularity never occurs in reality": a load-controlled physical system snaps and keeps a bounded tangent (Proposition 3), and the divergent $\epsilon^{-4/3}$ rate is felt only by an algorithm that suppresses the snap and tracks the unstable branch through the instability (Theorem 2). The spatial localisation below applies to that branch.

The maximum is, however, achieved only in a shrinking zone, and we can now identify the localisation exponent exactly. The relevant geometry is the *collapse locus* of the perfect field.

Lemma 11 (Transversal collapse: $s = 1$)**.** *On the perfect (unphysical) continuation of the fundamental path beyond t_c, let $\Sigma_c \subset B_\rho$ be the collapse locus on which the smallest stretch vanishes, $\lambda_{\min} = 0$ (equivalently $J = 0$); it is generically of codimension one. At any point of Σ_c where*

$\lambda_{\min}$ *is a simple singular value of* $\boldsymbol{F}$*,* $\lambda_{\min}$ *is a smooth function of* $\boldsymbol{x}$*, and generically it vanishes* transversally*,*

$$\lambda_{\min}^{\text{field}}(\boldsymbol{x}) = \kappa_* \ \mathrm{dist}(\boldsymbol{x}, \Sigma_c) + \mathcal{O}\big(\,\mathrm{dist}^2\,\big), \qquad \kappa_* > 0. \tag{61}$$

Hence the localisation exponent is $s = 1$*. (A non-generic tangential contact,* $\nabla\lambda_{\min} = 0$ *on* Σ_c*, would give* $s \geq 2$*.)*

Remark 11 (Extended arc versus point vertex). The codimension-one hypothesis is generic but not forced. In a compressive mixed corner the most severe contraction is frequently attained *exactly at the vertex* $r = 0$, in which case the collapse locus degenerates to the single point $\Sigma_c = \{0\}$ — codimension *two* in $d = 2$. The localisation conclusion then only improves: the relieved floor $\lambda_{\min} \gtrsim \max\{\kappa_* r,\, c\,\epsilon^{2/3}\}$ now confines the high-stiffness set to a *disc* of radius $\ell(\epsilon) \sim \epsilon^{2/3}$ rather than a tube of fixed length, so its area is $\mathcal{O}(\ell(\epsilon)^2) = \mathcal{O}(\epsilon^{4/3})$ rather than $\mathcal{O}(\epsilon^{2/3})$, and the L^q-bound of Proposition 4 acquires the stronger tube contribution $\epsilon^{-4q/3}\,\ell(\epsilon)^2 = \epsilon^{\frac{4}{3}(1-q)}$, finite for every $q < 1$ instead of $q < \frac{1}{2}$. The extended-arc estimate below is therefore the conservative case; a point vertex is strictly more integrable.

Proof. Where the smallest singular value is simple it is an analytic function of the entries of $\boldsymbol{F}$, hence smooth in $\boldsymbol{x}$. A smooth scalar field vanishing on Σ_c has, by Taylor expansion, $\lambda_{\min} = \nabla\lambda_{\min} \cdot \boldsymbol{n}\ \mathrm{dist}(\boldsymbol{x}, \Sigma_c) + \mathcal{O}(\mathrm{dist}^2)$ with $\boldsymbol{n}$ the unit normal; transversality $\nabla\lambda_{\min} \neq 0$ (the generic case, by Sard's theorem) gives (61) with $\kappa_* = |\nabla\lambda_{\min}| > 0$. □ □

On the imperfect branch the relief of (55) imposes the floor $\lambda_{\min}(\boldsymbol{x}, \epsilon) \gtrsim \max\{\kappa_*\,\mathrm{dist}(\boldsymbol{x}, \Sigma_c),\, c\,\epsilon^{2/3}\}$, so the high-stiffness zone is the tube of half-width $\ell(\epsilon)$ around Σ_c in which the floor is active, with (using $s = 1$)

$$\ell(\epsilon) \sim \frac{c}{\kappa_*}\,\epsilon^{2/3} \xrightarrow[\epsilon\to 0]{} 0. \tag{62}$$

Proposition 4 (Narrowness of the singular zone)**.** *With* $\ell(\epsilon)$ *as in* (62) *and* $s = 1$ *(Lemma 11), the set on which the tangent is near its maximum is a tube about* Σ_c *of area* $\mathcal{O}(|\Sigma_c|\,\ell(\epsilon)) = \mathcal{O}(\epsilon^{2/3}) \to 0$*, and the integrated stiffness obeys*

$$\int_{B_\rho} \|\mathbb{L}(\boldsymbol{x}, \epsilon)\|^q \, dV \ \leq\ C\Big(1 + \epsilon^{-4q/3}\,\ell(\epsilon)\,\log^q \tfrac{1}{\epsilon}\Big) \ =\ C\Big(1 + \epsilon^{\frac{2}{3}(1-2q)} \log^q \tfrac{1}{\epsilon}\Big), \tag{63}$$

which stays bounded as $\epsilon \to 0$ *for every* $q < \frac{1}{2}$*. Thus, although the pointwise tangent diverges like* $\epsilon^{-4/3}$*, its* L^q*-mass over the corner remains finite for all* $q < \frac{1}{2}$*: the blow-up is genuinely localised to a tube of vanishing area, and for every* $\epsilon > 0$ *the regularised problem is well posed.*

Proof. Split B_ρ into the tube $T_\epsilon = \{\mathrm{dist}(\cdot, \Sigma_c) < \ell(\epsilon)\}$ and its complement. On T_ϵ, $\|\mathbb{L}\| \sim \epsilon^{-4/3}\log\frac{1}{\epsilon}$ (Theorem 2) and $|T_\epsilon| \sim |\Sigma_c|\ell(\epsilon) \sim \epsilon^{2/3}$, giving the displayed contribution. Off T_ϵ, using the transversal coordinate $\varrho = \mathrm{dist}(\cdot, \Sigma_c)$ and $\lambda_{\min} \sim \kappa_*\varrho$, one has $\|\mathbb{L}\| \sim \varrho^{-2}\log\frac{1}{\varrho}$, and $\int_{\ell(\epsilon)}^{\rho} \varrho^{-2q}\log^q\frac{1}{\varrho}\,d\varrho$ converges as $\ell(\epsilon) \to 0$ precisely when $2q < 1$. Both regimes are controlled iff $q < \frac{1}{2}$. □ □

4.7 Discussion

The structure of the result is the one anticipated physically, and it splits cleanly along the stability of the branch followed. The idealised noiseless problem ($\epsilon = 0$) is singular—the tangent *norm* is unbounded ($\|\mathbb{L}\| \to \infty$), even though the corner remains strongly elliptic pointwise (Lemma 5)—but it is a measure-zero idealisation. Because the corner instability is subcritical (the creasing analogy, Assumption 7), a load-controlled physical system never traverses the singular continuation: it stays on the compressed branch and *snaps* to a finite-amplitude crease at a load reduced by only $\mathcal{O}(\epsilon^{2/3})$ (Proposition 2), keeping a bounded tangent throughout (Proposition 3). The much-discussed divergence is therefore not seen by the physical system at all; it is the worst case for an algorithm—displacement or arc-length control, numerical continuation—that suppresses the snap and tracks the *unstable* branch through the instability. On that branch the determinant is still lifted off zero (Lemma 9, with the asymptotically volume-preserving character of the mode posited by Assumption 8 and confirmed numerically, Remark 10), the tangent diverges only algebraically, $\epsilon^{-4/3}$ up to a logarithm (Theorem 2), and the singular character is confined to a tube of width $\ell(\epsilon) \sim \epsilon^{2/3} \to 0$ about the collapse locus, remaining L^q-integrable for all $q < \frac{1}{2}$ (Lemma 11, Proposition 4). That algebraic rate is the mirror image—through the inverse channel—of the benign tensile growth of §3.4; the same logarithm, traceable to the barrier $f_{\log}$, appears in both. The headline is therefore analytical rather than computational: a compressed clamped–free corner harbours a genuine structural bifurcation — the corner-localised analogue of the classical *creasing* instability of compressed elastomers [1, 3], now seated at a geometric Dirichlet–Neumann junction rather than on a flat surface — whose apparent tangent catastrophe is, on every physical branch, regularised by the kinematic relief (Assumption 8, numerically supported). That the phenomenon first surfaced as ill-conditioning in a numerical computation is incidental to its nature; the conditioning degradation is a corollary, not the subject. It does carry a concrete computational reading — a solver that suppresses the snap and tracks the unstable branch sees the conditioning degrade as $\epsilon^{-4/3}$ and should either admit the snap or refine inside the tube (62), the load-control reduction (44) telling it when the snap is due — but this is one consequence among others, not the result. Section 5 pursues the sharper, opposite direction: fixing a concrete material and geometry, it discharges the abstract hypotheses to theorems and shows, through a blow-up reduction, that the entire phenomenon localises to a *universal* model wedge governed only by the Poisson ratio and a single stress-intensity amplitude. The reduction to scalar imperfect-bifurcation form follows the classical theory [2, 4, 5].

5 A concrete realization: the clamped–free corner under compression

The preceding analysis is deliberately abstract: it isolates the structural properties of the energy (Assumptions 1–5) and of the loading (the hypotheses of §4) that the argument actually uses, without committing to a domain. We now fix a concrete material and a concrete family of geometries and ask how much of that hypothesis list can be *discharged* rather than assumed. The material is the compressible neo-Hookean energy (7) with the barrier penalty $f_{\log}$. The geometry is a body whose boundary carries a clamped (Dirichlet) face meeting a traction-free (Neumann) face at a right angle, under a compressive load.

The abstract development of §3–§4 is already dimension-independent: it analyses the transverse problem at the codimension-two Dirichlet–Neumann edge Σ in $\mathbb{R}^d$. Here we instantiate it on concrete bodies and ask how much of that hypothesis list can be *discharged* rather than assumed.

5.1 Geometry and the singular set

We consider two model bodies in $\Omega \subset \mathbb{R}^d$.

- **(H) Half-ball.** $\Omega = \{\, \boldsymbol{x} \in \mathbb{R}^d : |\boldsymbol{x}| < 1,\ x_d > 0 \,\}$. The flat base $\Gamma_D = \{x_d = 0\} \cap \partial\Omega$ is clamped; the spherical cap $\Gamma_N = \{|\boldsymbol{x}| = 1\} \cap \partial\Omega$ is traction-free.

- **(C) Box.** $\Omega = (-1,1)^{d-1} \times (0,1)$. The base $\Gamma_D = \{x_d = 0\}$ is clamped; the remaining faces are traction-free.

In both cases the load is a monotone compression transmitted through Γ_N (a pressure or an inward normal displacement on the cap, respectively the lateral/top faces), scaled by the pseudo-time $t \in [0,1]$ of §4. We carry two bodies not because they require two analyses — by the blow-up reduction of §5.2 they do not, see Remark 12 — but because they bracket the two situations one meets in practice: a globally smooth edge (H) and a polyhedral one with vertices (C).

The *singular set* $\Sigma \subset \partial\Omega$ is the locus where $\overline{\Gamma_D}$ meets $\overline{\Gamma_N}$:

- for (H), $\Sigma = \{|\boldsymbol{x}| = 1,\ x_d = 0\} \cong S^{d-2}$, a *smooth* codimension-two submanifold; the cap is tangent to the vertical at the equator, so it meets the horizontal base at a right angle;
- for (C), Σ is the union of the bottom edges $\{x_d = 0,\ x_i = \pm 1\}$, again right-angle Dirichlet–Neumann edges, *together with* the bottom vertices, where d faces meet.

Fix $\boldsymbol{x}_0 \in \Sigma$ away from any vertex and split $\mathbb{R}^d = T_{\boldsymbol{x}_0}\Sigma \oplus \Pi$, with Π the two-plane normal to Σ. In the slice Π the boundary reduces to a Dirichlet ray and a Neumann ray meeting at $90°$: *exactly* the planar junction of §3.5. We write B_ρ for a tubular neighbourhood of Σ of radius ρ and use (r,θ) for polar coordinates in Π as before.

Remark 12 (The two bodies are one edge analysis). The blow-up reduction (Proposition 5) makes precise that (H) and (C) are *not* two cases but one: at any non-vertex point of Σ the transverse problem is the same planar right-angle clamped–free wedge, with the *same* exponent α, angular profile $\boldsymbol{\Phi}$ and reduced data, the bodies differing only through the scalar stress-intensity $\hat{K}$ (a point in the phase diagram of Remark 13) and through edge curvature, which enters only at the subleading order $\mathcal{O}(\lambda^g)$. The *single* genuine distinction is topological. For (H) the singular set $\Sigma \cong S^{d-2}$ is a smooth closed manifold with no vertices, so the singularity is a pure edge everywhere on Σ and the analysis holds verbatim at every point. For (C) the edge interiors are identical, but the bottom vertices, where d faces meet, carry an additional, generally stronger, polyhedral-vertex singularity [7] outside the scope of the edge analysis; one either rounds the vertices or restricts B_ρ to the edge interior. We therefore state the dimension-independent results once, read on (H) as the clean smooth-edge representative, and note that they hold verbatim on the edge interiors of (C).

5.2 The blow-up principle: reduction to the model wedge

The analysis of §3–§4 works on a tubular neighbourhood B_ρ of Σ cut from a *specific* body. The informal reason the two bodies of §5.1 share the same corner physics is that the singularity is localised at Σ: the leading data at the edge cannot feel the far field except through the single amplitude K. We now make this precise as a blow-up (tangent-cone) reduction, which is what licenses replacing "numerics on (H) and (C)" by "numerics on one universal model."

Fix $\boldsymbol{x}_0 \in \Sigma$ away from any vertex and let $W := \Pi_W \times T_{\boldsymbol{x}_0}\Sigma$ be the *tangent wedge*, with $\Pi_W = \{(r,\theta) : r > 0,\ 0 < \theta < \frac{\pi}{2}\}$ the right-angle Dirichlet–Neumann quarter-plane in the normal two-plane Π. For $\lambda \in (0,1]$ introduce the rescaling $\boldsymbol{\xi} \mapsto \boldsymbol{x}_0 + \lambda\boldsymbol{\xi}$, under which $B_{\lambda R}(\boldsymbol{x}_0)$ maps to $B_R(\boldsymbol{0}) \cap W$ and $\Omega_\lambda := \lambda^{-1}(\Omega - \boldsymbol{x}_0) \to W$ as $\lambda \to 0$. Write the fundamental displacement on the loaded path as $\boldsymbol{u}_0 = K\, r^\alpha \boldsymbol{\Phi}(\theta) + \boldsymbol{u}_0^\sharp$, where by the Kondratiev expansion the remainder collects the next exponents, $\boldsymbol{u}_0^\sharp = \mathcal{O}(r^{\alpha'})$ with $\alpha' > \alpha$ the second singular exponent of the pencil, and set the *spectral gap* $g := \alpha' - \alpha > 0$. Define the rescaled displacement, tangent and operator

$$\hat{\boldsymbol{u}}_\lambda(\boldsymbol{\xi}) := \lambda^{-\alpha}\big(\boldsymbol{u}_0(\boldsymbol{x}_0 + \lambda\boldsymbol{\xi}) - \boldsymbol{u}_0(\boldsymbol{x}_0)\big), \qquad \widehat{\mathbb{L}}_\lambda(\boldsymbol{\xi}) := \mathbb{L}\big(\boldsymbol{F}_0(\boldsymbol{x}_0 + \lambda\boldsymbol{\xi})\big), \qquad \widehat{\mathcal{L}}_\lambda := \partial_{\boldsymbol{u}} G\big(\boldsymbol{u}_0, \cdot, 0\big)\big|_{\boldsymbol{x}_0 + \lambda\boldsymbol{\xi}}. \tag{64}$$

Proposition 5 (Blow-up reduction to the model wedge)**.** *Let $\boldsymbol{x}_0 \in \Sigma$ be a non-vertex point and let $\beta_\star$ be the Fredholm weight of §3. Then, as $\lambda \to 0$:*

(i) Kinematics. *$\hat{\boldsymbol{u}}_\lambda \to \hat{\boldsymbol{u}}_W := K\, r^\alpha \boldsymbol{\Phi}(\theta)$ in $V^{1,2}_{\beta_\star}(B_R \cap W)$ for every $R > 0$, with rate $\|\hat{\boldsymbol{u}}_\lambda - \hat{\boldsymbol{u}}_W\|_{V^{1,2}_{\beta_\star}(B_R\cap W)} = \mathcal{O}(\lambda^g)$. The limit solves the homogeneous linear pencil on W and is fixed by the wedge geometry up to the scalar K.*

(ii) Operator convergence away from the tip; common vertex pencil. *On every annulus $\{0 < r_0 \leq r \leq R\} \cap W$ bounded away from the vertex, the coefficients of $\widehat{\mathcal{L}}_\lambda$ converge in C^0, with rate $\mathcal{O}(\lambda^g)$, to those of the model-wedge operator $\mathcal{L}_W$ obtained by freezing along the self-similar field $\hat{\boldsymbol{u}}_W$. The leading singular exponent α and angular profile $\boldsymbol{\Phi}$ are* not *inferred from this annular convergence (they are pencil/vertex data, which annular C^0 control cannot reach); rather, they are* identical *for $\widehat{\mathcal{L}}_\lambda$, $\mathcal{L}_W$ and the linearised-elasticity operator alike, because all three share the* same *leading vertex pencil. Indeed at the tip the nonlinear correction to the principal symbol is subordinate ($o(r^{-\delta})$ for every δ, Lemma 2), so by Theorem 1 the corner pencil — and hence its eigenvalues α and eigenprofiles $\boldsymbol{\Phi}$ — coincides with that of constant-coefficient linear elasticity, unchanged by the nonlinearity and independent of λ. Thus only the dimensionless amplitude K (equivalently the normalised stress-intensity $\hat{K}$ of Remark 13) survives from the global problem. We emphasise what is* not *asserted: the away-from-tip convergence governs only the lower-order (non-principal) coefficients; it does not control the operator in a weighted norm up to $r = 0$, and the agreement of the full Lyapunov–Schmidt reduced data is a leading-order statement rather than a quantitative $\mathcal{O}(\lambda^g)$ spectral-perturbation theorem, a uniform-to-the-tip estimate requiring the inner vertex theory left open in Remark 2.*

(iii) Dimensional reduction $d \geq 3 \Rightarrow d = 2$. *Since Σ is smooth and the coefficients are constant along $T_{\boldsymbol{x}_0}\Sigma$ in the limit, the Fourier transform in the edge variable $\boldsymbol{s} \in T_{\boldsymbol{x}_0}\Sigma$ decomposes $\mathcal{L}_W$ into a one-parameter family $\{\mathcal{L}_W(\boldsymbol{\zeta})\}_{\boldsymbol{\zeta}\in\widehat{T_{\boldsymbol{x}_0}\Sigma}}$ of operators on the planar quarter-plane Π_W, each the transverse two-dimensional pencil of* (13)*; the leading edge exponent is carried by the zero wavenumber $\boldsymbol{\zeta} = \boldsymbol{0}$. The model problem is therefore genuinely two-dimensional.*

Proof. Statement (i) is the leading-term extraction of the Kondratiev–Maz'ya–Rossmann expansion [6, 7]: on the Fredholm weight $\beta_\star$ the solution admits $\boldsymbol{u}_0 = Kr^\alpha\boldsymbol{\Phi} + \mathcal{O}(r^{\alpha'})$, and the rescaling (64) maps the leading term to the λ-independent field $Kr^\alpha\boldsymbol{\Phi}$ exactly while shrinking the remainder by $\lambda^{\alpha'-\alpha} = \lambda^g$ in the homogeneous weighted norm. Statement (ii) follows because $\mathbb{L}$ is a smooth (algebraic) function of $\boldsymbol{F}_0 = \boldsymbol{I} + \nabla\boldsymbol{u}_0$ and $\nabla\hat{\boldsymbol{u}}_\lambda \to \nabla\hat{\boldsymbol{u}}_W$ off the tip by (i); freezing along the self-similar limit defines $\mathcal{L}_W$, and the coefficient error inherits the $\mathcal{O}(\lambda^g)$ rate of (i) on every annulus $\{r_0 \leq r \leq R\}$ bounded away from $r = 0$. The leading exponent and angular profile are obtained not from this annular convergence but from the principal-symbol structure at the vertex: by Lemma 2 the nonlinear correction to the principal symbol is subordinate (relative bound 0), so by Theorem 1 the corner pencils of $\widehat{\mathcal{L}}_\lambda$ and of $\mathcal{L}_W$ both coincide with that of constant-coefficient linear elasticity at $\boldsymbol{x}_0$; their eigenvalues α and eigenprofiles $\boldsymbol{\Phi}$ are therefore equal (not merely close), independently of λ. We do *not* claim a weighted operator-norm estimate uniform up to $r = 0$: the tip contributes through the weight, not through the body, and its quantitative control is the open inner problem of Remark 2; accordingly the agreement of the full reduced bifurcation data is asserted only at leading order. Statement (iii) is the standard edge-to-corner reduction: translation invariance along $T_{\boldsymbol{x}_0}\Sigma$ in the limit diagonalises $\mathcal{L}_W$ under the partial Fourier transform, and each fibre is the planar pencil whose smallest root is α, attained at $\boldsymbol{\zeta} = \boldsymbol{0}$ (the cited edge calculus). □ □

Remark 13 (The one surviving global datum, and what it costs). Proposition 5 isolates exactly what does and does not localise. The exponent α, the angular profile $\boldsymbol{\Phi}$, and the wedge operator $\mathcal{L}_W$ are *universal*: functions of the opening (90°) and the Poisson ratio alone. The only trace of the finite body is the scalar K, conveniently made dimensionless as $\hat{K} := K\,\rho^{\alpha-1}/\mu_e$ (the normalised stress-intensity over the neighbourhood scale). Thus every quantity in the corner reduces to a computation on $\mathcal{L}_W$ as a function of $(\nu, \hat{K})$ — a two-parameter *phase diagram*, not a per-body spot check. Whether a given quantity depends on $\hat{K}$ at all, or only on (ν), is decided by a homogeneity count against α, carried out for the two open conditions in §5.9 (condition (BC), which needs only ν) and §5.7 (the cubic b, which retains the $\hat{K}$-dependence). This is the precise, and sharper, replacement for "the singularity is localised to the corner."

5.3 Tier 1: the material assumptions are already discharged

For the neo-Hookean energy (7) with $f_{\log}$, the standing assumptions of §3 hold with explicit constants and require no information about the domain: Assumption 1 (regularity, objectivity, and the collapse barrier $f_{\log}(J) \to \infty$), Assumption 2 (the inverse-channel bound, with $\Theta = \Theta_{\log}$ of (23)), Assumption 3 and the sharper Lemma 5 (strong ellipticity with constant $\geq \mu_e$ throughout compression), and Assumption 5 (logarithmic growth, $p = 1$) are all established in Lemma 1, Remark 1 and §3.4. None of these proofs uses $d = 2$: the tangent formula (10), the rank-one identity (18), and the growth functions (23) are algebraic identities in $\mathrm{GL}^+(d)$. Thus the entire material side of the hypothesis list is closed for the concrete model, in every dimension. What remains are the hypotheses on the *solution and loading*, which is where the geometry enters.

5.4 The edge exponent and the scenario it selects

By the edge-singularity calculus [6, 7], the leading singular exponents of the linearised elasticity operator along the smooth edge Σ coincide with those of the transverse two-dimensional pencil in Π, modulated smoothly along Σ. That transverse pencil is the planar right-angle Dirichlet–Neumann corner of (13); its leading exponent $\alpha \in (0, 1)$ is the smallest root of the corresponding clamped–free Williams secular equation and depends only on the Poisson ratio. Consequently the kinematic blow-up (14), $\boldsymbol{F} = \boldsymbol{I} + \mathcal{O}(r^{\alpha-1})$, holds with a *computable* exponent, and by Theorem 1 the nonlinear tangent inherits exactly this α in the benign regime — on the annulus outside the dilation core of Lemma 3, the core being absent for mild loading or a globally convex-volumetric barrier.

Which scenario of §3 is realized is governed by the sign of the dominant eigenvalue of $\nabla \boldsymbol{u}$ at the edge, equivalently by the sign of the stress-intensity factor K in (13). This is a global quantity, set by the far field, and is not determined by the local pencil alone. The compressive load we impose selects $K < 0$: the material is pushed into the corner, the dominant eigenvalue of $\nabla \boldsymbol{u}$ is negative, and Scenario B (§3.6) is the relevant branch. The quantitative form of "compressive enough" is the hypothesis (SC) below; it simultaneously selects Scenario B and drives the structural instability of §5.5. The complementary tensile load ($K > 0$) selects Scenario A, where (NC) holds and Theorem 1 applies; for the present geometry under compression that case does not arise near the edge.

5.5 Existence of the critical load via a localized surface instability

The one genuinely non-elementary ingredient of §4 was hypothesis (ii) of Lemma 6: the existence of a structurally destabilising field, $Q_{t_*}(\boldsymbol{v}_*) < 0$, which we there declined to establish and flagged as the corner counterpart of the Biot/creasing instability. For the concrete clamped–free geometry we now *prove* it, by reduction to the surface instability of the compressed half-space.

The free face Γ_N under compression is, locally, a traction-free surface compressed parallel to itself — the exact configuration of Biot's classical analysis [1]. A homogeneously compressed neo-Hookean half-space loses stability to a surface mode once the surface-parallel stretch falls below a threshold $\lambda_\star \in (0,1)$ (for the incompressible model $\lambda_\star \approx 0.544$; for the compressible energy (7) the threshold is the corresponding root of the surface secular equation, still in $(0,1)$ [3]). We package the loading requirement as a single hypothesis.

Assumption 10 (Surface compression reaches threshold, (SC))**.** There exist $t_* \in (0,1]$ and an interior point $P \in \Gamma_N \cap B_\rho$, at positive distance from both Σ and $\overline{\Gamma_D}$, such that the fundamental state compresses the free face below the half-space threshold there: the surface-parallel principal stretch of $\boldsymbol{F}_0(t_*)$ at P satisfies $\lambda_\parallel\big(\boldsymbol{F}_0(t_*); P\big) < \lambda_\star$.

(SC) is an input on the *loading*, in the same spirit as the monotonicity condition (M) of Lemma 7: it says only that the prescribed compression is strong enough that the free surface dips below its own buckling threshold somewhere near the edge. Since t merely normalises the load amplitude, (SC) can always be met by loading hard enough, provided the material does not fail elsewhere first.

Lemma 12 (Localized surface instability $\Rightarrow$ structural instability)**.** *Under Assumption 10, hypothesis (ii) of Lemma 6 holds: there is $\boldsymbol{v}_* \in V$, supported in B_ρ and vanishing on Γ_D, with*

$$Q_{t_*}(\boldsymbol{v}_*) = \int_{B_\rho} \nabla \boldsymbol{v}_* : \mathbb{L}\big(\boldsymbol{F}_0(t_*)\big) : \nabla \boldsymbol{v}_* \, dV < 0. \tag{65}$$

Consequently, with $\mu_1(0) > 0$ from Lemma 6(i), there is a first critical load $t_c \in (0, t_*]$ with $\mu_1(t_c) = 0$.

Proof. Work in coordinates centred at P: let $n \geq 0$ be the depth into the body along the inward normal to Γ_N and $\boldsymbol{s} \in \mathbb{R}^{d-1}$ the surface-tangent coordinates, so that near P the body is the half-space $\{n > 0\}$ and $\boldsymbol{F}_0(t_*)$ is C^1 with value $\boldsymbol{F}_P := \boldsymbol{F}_0(t_*; P)$ whose surface-parallel stretch is $\lambda_\parallel < \lambda_\star$.

Step 1: the homogeneous unstable mode. Freeze the coefficients at $\boldsymbol{F}_P$ and consider the constant-tensor form $Q_0(\boldsymbol{v}) = \int_{\{n>0\}} \nabla \boldsymbol{v} : \mathbb{L}(\boldsymbol{F}_P) : \nabla \boldsymbol{v} \, dV$ on the half-space, with $\boldsymbol{v}$ traction-free at $n = 0$. Because $\lambda_\parallel < \lambda_\star$, the half-space is past its surface-instability threshold [1, 3]: for each surface wavevector $\boldsymbol{k} \in \mathbb{R}^{d-1}$, $k := |\boldsymbol{k}|$, there is a surface mode

$$\boldsymbol{v}_{\boldsymbol{k}}(\boldsymbol{s}, n) = \mathrm{Re}\left[\, \hat{\boldsymbol{v}}(kn)\, e^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{s}} \,\right], \qquad |\hat{\boldsymbol{v}}(\zeta)| \leq Ce^{-\beta\zeta} \ (\beta > 0), \tag{66}$$

decaying exponentially into the body on the depth scale $1/k$. Since $\mathbb{L}(\boldsymbol{F}_P)$ is constant, the rescaling $(\boldsymbol{s}, n) \mapsto k^{-1}(\boldsymbol{s}, n)$ shows that over a surface window W of area $|W|$,

$$\int_{\{n>0\}} \mathbf{1}_W \, \nabla \boldsymbol{v}_{\boldsymbol{k}} : \mathbb{L}(\boldsymbol{F}_P) : \nabla \boldsymbol{v}_{\boldsymbol{k}} \, dV = q(\lambda_\parallel)\, k\, |W|\, (1 + o(1)), \qquad q(\lambda_\parallel) < 0, \tag{67}$$

where q is the Biot secular form, strictly negative past threshold. Likewise $\int_{\{n>0\}} \mathbf{1}_W \, |\nabla \boldsymbol{v}_{\boldsymbol{k}}|^2 \sim k\,|W|$ and $\int_{\{n>0\}} \mathbf{1}_W \, |\boldsymbol{v}_{\boldsymbol{k}}|^2 \sim k^{-1}|W|$, the latter two by the same scaling.

Step 2: cut-off. Let χ be smooth, $\chi \equiv 1$ on $B(P, \delta/2)$, $\operatorname{supp}\chi \subset B(P, \delta)$, $|\nabla\chi| \leq C/\delta$, with δ small enough that $B(P, \delta) \cap \overline{\Omega}$ avoids both Σ and $\overline{\Gamma_D}$ and lies in B_ρ (possible since P is at positive distance from both). Set $\boldsymbol{v}_* := \chi\, \boldsymbol{v}_{\boldsymbol{k}}$. Then $\boldsymbol{v}_* \in V$ and vanishes on Γ_D (its support avoids $\overline{\Gamma_D}$); it is unconstrained on the free face, as required.

Step 3: error control. Write $\nabla \boldsymbol{v}_* = \chi \nabla \boldsymbol{v}_{\boldsymbol{k}} + (\nabla\chi) \otimes \boldsymbol{v}_{\boldsymbol{k}}$ and expand (65). The effective window is $W \sim B(P, \delta) \cap \Gamma_N$, of area $|W| \sim \delta^{d-1}$, and the decay confines the mass to depth $1/k$. Three errors arise.

(a) *Coefficient variation:* replacing $\mathbb{L}(\boldsymbol{F}_0(\boldsymbol{x}))$ by $\mathbb{L}(\boldsymbol{F}_P)$ costs $\le \sup_{B(P,\delta)} \|\mathbb{L}(\boldsymbol{F}_0) - \mathbb{L}(\boldsymbol{F}_P)\| \int \chi^2 |\nabla \boldsymbol{v}_{\boldsymbol{k}}|^2 \le C\delta \cdot k\delta^{d-1} = Ck\delta^d$ (by C^1 regularity of $\boldsymbol{F}_0$ at P).

(b) *Gradient commutator:* the cross term and $|\nabla\chi|^2 |\boldsymbol{v}_{\boldsymbol{k}}|^2$ term are bounded by $C\big(\delta^{-1}\int_W |\boldsymbol{v}_{\boldsymbol{k}}||\nabla \boldsymbol{v}_{\boldsymbol{k}}| + \delta^{-2}\int_W |\boldsymbol{v}_{\boldsymbol{k}}|^2\big) \le C\big(\delta^{d-2} + \delta^{d-3}k^{-1}\big)$.

(c) *Window vs. cut-off:* replacing $\mathbf{1}_W$ by χ^2 in (67) changes the leading constant by $1 + o(1)$ only.

The leading term is $q(\lambda_\parallel)\, k\, \delta^{d-1} < 0$. The errors (a)–(b), relative to it, are $\mathcal{O}(\delta)$, $\mathcal{O}((k\delta)^{-1})$ and $\mathcal{O}((k\delta)^{-2})$. Choosing $\delta = k^{-1/2}$ sends $\delta \to 0$ and $k\delta = k^{1/2} \to \infty$, so all relative errors vanish as $k \to \infty$. Hence for k large enough,

$$Q_{t_*}(\boldsymbol{v}_*) = q(\lambda_\parallel)\, k\, \delta^{d-1}\, (1 + o(1)) < 0,$$

which is (65). The final statement is Lemma 6 with hypothesis (ii) now supplied. □ □

Remark 14 (What this closes). Lemma 12 converts the existence of the critical load t_c — previously conditional on the unproved hypothesis (ii) — into a *theorem* for the concrete clamped–free geometry, contingent only on the loading hypothesis (SC). What the construction delivers, and what it does not, should be stated precisely. It exhibits a single admissible field $\boldsymbol{v}_*$ with $Q_{t_*}(\boldsymbol{v}_*) < 0$, which by Lemma 6 is all that is needed to force a crossing $\mu_1(t_c) = 0$ at some $t_c \in (0, t_*]$. The destabilising mechanism is the half-space surface (creasing) instability, seated at an interior free-face point P of the edge neighbourhood B_ρ and confined to it by the cut-off (66)–(67); the witness $\boldsymbol{v}_*$ is supported away from the tip $r = 0$ (indeed (SC) places P at positive distance from Σ), so the lemma certifies a surface instability *within* B_ρ, not a mode pinned at the corner vertex. This is the precise sense in which the corner bifurcation is "the corner analogue of creasing": the same free-face mechanism, activated near — not at — the edge. The location and shape of the genuine *critical* mode $\boldsymbol{\phi}$ at t_c is a separate question, settled by the attainment analysis of §5.9 (which gives $\boldsymbol{\phi} \in V$ supported in B_ρ), and the role of the edge proper enters quantitatively only through the graded pre-stress of Remark 16.

Remark 15 (On the threshold $\lambda_\star$: compressible vs. incompressible). The proof uses only the *existence* of a surface-instability threshold $\lambda_\star \in (0, 1)$, not its value. The familiar number $\lambda_\star \approx 0.544$ is Biot's result for the *incompressible* neo-Hookean half-space [1]; the present material (7) is compressible, and its threshold is the corresponding root of the compressible surface secular equation, which depends on the ratio λ_e/μ_e (equivalently the Poisson ratio) and tends to 0.544 in the incompressible limit $\lambda_e/\mu_e \to \infty$ [3]. This is the same instability with a material-dependent threshold, not a discrepancy: Lemma 12 *reduces* the corner instability to the half-space one and thereby agrees with the creasing literature rather than competing with it. Accordingly we keep $\lambda_\star$ symbolic throughout.

5.6 Symmetry: simplicity of the mode and the vanishing quadratic

Both model bodies of §5.1 possess reflection symmetries that the fundamental compressive state inherits: (H) is invariant under every orthogonal map fixing the x_d-axis, and (C) under each coordinate reflection $x_i \mapsto -x_i$ ($i < d$). Let R be such a reflection and $\mathcal{R}$ its action on displacement fields. Because $\boldsymbol{F}_0(t)$ and hence $\mathbb{L}(\boldsymbol{F}_0(t))$ are R-invariant, the second-variation operator $\mathcal{L}(t)$ commutes with $\mathcal{R}$, so its eigenspaces split into the symmetric ($\mathcal{R} = +1$) and antisymmetric ($\mathcal{R} = -1$) sectors. This yields, for the concrete geometry, two of the ingredients that §4 had to posit:

- *Simplicity (part of Assumption 6).* The lowest eigenvalue within a fixed symmetry sector is generically simple; the genericity is the standard one for self-adjoint operators commuting with a finite symmetry group. This reduces simplicity to two residual genericity statements — that the crossing eigenvalue is the ground state of its sector, and that no accidental degeneracy across sectors occurs at t_c — rather than discharging it unconditionally. With that caveat, simplicity need not be assumed as an independent spectral postulate, and only the *attainment* clause (the borderline-elliptic point of Remark 6) requires the separate analysis of §5.9.

- *Vanishing quadratic coefficient (part of Assumption 7).* The reduced potential of Proposition 1 inherits the $\mathcal{R}$-symmetry of the perfect state. If $\boldsymbol{\phi}$ lies in the antisymmetric sector, $w \mapsto -w$ is a symmetry of the reduced equation and the quadratic coefficient vanishes. For the neo-Hookean model this is made fully explicit in Lemma 13: c_2 is odd in the first-order volume change D_1, which R flips, so $c_2 = 0$ exactly. The relevant reflection R is the lateral, *face-preserving* symmetry of the model bodies of §5.1, *not* the corner bisector (which would interchange Γ_D and Γ_N and is no symmetry of the mixed problem).

Lemma 13 (Neo-Hookean reduced coefficients are volumetric; the quadratic vanishes)**.** *For the neo-Hookean energy* (7) *the deviatoric term* $\frac{\mu_e}{2}(\operatorname{tr}\boldsymbol{F}^T\boldsymbol{F} - d)$ *is quadratic in* $\boldsymbol{F}$*, so its third and higher Fréchet derivatives vanish; hence* every *reduced bifurcation coefficient of order* ≥ 3 *is purely volumetric, determined by* f *alone. Writing* $J(w) = \det(\boldsymbol{F}_0 + w\nabla\boldsymbol{\phi}) = \sum_k D_k\, w^k$ *with* D_k *the mixed discriminants of Lemma 9* ($D_1 = \operatorname{cof}\boldsymbol{F}_0 : \nabla\boldsymbol{\phi}$, $D_2 = d_\phi$, $D_k = 0$ *for* $k > d$) *and* $f^{(k)} = f^{(k)}(J_0)$*, the single-mode coefficients are*

$$c_2 = \tfrac{1}{2}\int_{B_\rho} \left[f''' D_1^3 + 6 f'' D_1 D_2 + 6 f' D_3\right] dV, \tag{68}$$

$$b_{\mathrm{dir}} = \tfrac{1}{6}\int_{B_\rho} \left[f'''' D_1^4 + 12 f''' D_1^2 D_2 + 12 f'' D_2^2 + 24 f'' D_1 D_3 + 24 f' D_4\right] dV. \tag{69}$$

Two consequences hold for the concrete geometry, whose fundamental state is R*-symmetric.*

(i) The quadratic vanishes, exactly and without correction. *As a* third-*order quantity* c_2 *receives no orthogonal-mode (Lyapunov–Schmidt) feedback — that correction first enters at fourth order — so* (68) *is the full* c_2*. The key parity fact is that each mixed discriminant* $D_k = \frac{1}{k!}\partial_w^k \det(\boldsymbol{F}_0 + w\nabla\boldsymbol{\phi})|_0$ *is exactly the degree-*k *part of the determinant in* $\nabla\boldsymbol{\phi}$*, hence carries* k *factors of* $\nabla\boldsymbol{\phi}$ *and one factor of an* R*-even cofactor of* $\boldsymbol{F}_0$ *per term. For an* R*-antisymmetric mode* $\nabla\boldsymbol{\phi}$ *reverses sign under the reflection (composed with the change of variables* $\boldsymbol{x} \mapsto R\boldsymbol{x}$ *on the* R*-symmetric domain* B_ρ*), so* D_k *acquires the parity* $(-1)^k$*:* D_1 *and* D_3 *are* R*-odd,* D_2 *is* R*-even, while* $f^{(k)}(J_0)$ *is* R*-even (the base state is* R*-invariant). Each term of* (68) *thus carries an* odd *total number of* $\nabla\boldsymbol{\phi}$ *factors —* D_1^3 *(three),* $D_1 D_2$ *(three), and* D_3 *(three) — and is therefore* R*-odd; the integrand is odd over the* R*-symmetric* B_ρ *and* $c_2 = 0$*. (In* $d = 2$ *the term* $6f' D_3$ *is moreover identically zero, since* $D_k = 0$ *for* $k > d$*; the parity argument is what disposes of it in* $d \geq 3$*.) This* proves *the vanishing-quadratic clause of Assumption 7.*

(ii) The frozen cubic and quintic are sign-definite, and adverse to subcriticality. *With* D_1 *asymptotically small (Assumption 8), the leading frozen parts are* $b_{\mathrm{dir}} = 2\int_{B_\rho} f''(J_0)\, d_\phi^2\, dV$ *and* $d_{\mathrm{dir}} = \int_{B_\rho} f'''(J_0)\, d_\phi^3\, dV$*. For the logarithmic penalty* $f''_{\log}(J_0) = (\mu_e + 2\lambda_e - 2\lambda_e \log J_0)/J_0^2 > 0$ *and* $f'''_{\log}(J_0) = (4\lambda_e \log J_0 - 6\lambda_e - 2\mu_e)/J_0^3 < 0$ *throughout compression* $J_0 < 1$*. The cubic part carries* $d_\phi^2 \geq 0$*, so* $b_{\mathrm{dir}} > 0$ *follows from* $f''_{\log} > 0$ alone*, with no sign hypothesis on* d_ϕ *(this is the only frozen sign the argument needs). The quintic part carries the sign-indefinite*

d_ϕ^3*; on the relief-oriented region* $d_\phi > 0$ *((72) below) and* $f'''_{\log} < 0$ *give* $d_{\rm dir} < 0$ *there — the* opposite *of the subcritical pattern* $b < 0 < d$*, so the frozen quintic too is adverse where the relief acts, and subcriticality is decided by the feedback, not by either frozen term.*

Proof. Expand $\Psi(\boldsymbol{F}_0+w\nabla\boldsymbol{\phi}) = \frac{\mu_e}{2}(\operatorname{tr}\boldsymbol{F}^T\boldsymbol{F}-d)+f(J(w))$ in w. The deviatoric part is a polynomial of degree two in w, so it contributes only to the coefficients of w^0, w^1, w^2; all coefficients of order ≥ 3 come from $f(J(w))$, whose w-derivatives are given by Faà di Bruno's formula in the J-derivatives $f^{(k)}$ and the discriminants $D_k = \frac{1}{k!}\partial_w^k \det(\boldsymbol{F}_0+w\nabla\boldsymbol{\phi})|_0$. Reading off the third and fourth derivatives gives (68)–(69); statements (i)–(ii) are then as stated, using $f''_{\log} > 0$, $f'''_{\log} < 0$ on $J_0 < 1$. □ □

What symmetry fixes is thus the quadratic coefficient; what it does *not* fix is the sign of the cubic b and quintic d. Indeed the explicit single-mode parts (69) point *against* subcriticality ($b_{\rm dir} > 0$, $d_{\rm dir} < 0$ by Lemma 13(ii)), so the subcritical pattern $b < 0 < d$ rests entirely on the geometry-dependent orthogonal-mode feedback — the irreducible core, which we settle numerically rather than by analogy (Remark 8). Proposition 6 below explains, by a homogeneity count against the edge exponent α, *why* this feedback cannot be reduced to a universal corner number.

5.7 The homogeneity count for the cubic

The blow-up reduction of §5.2 promised a dichotomy: a corner quantity localises to a universal model-wedge number only when its defining integral is dominated by the tip. We now carry out the count for the cubic coefficient and find that, for the right-angle clamped–free pencil, it sits on the *non*-localising side — which is exactly why the sign of b is a genuine, $\hat{K}$-dependent computation rather than an inheritance from flat creasing.

Near the edge the critical mode $\boldsymbol{\phi} \in V$ inherits the admissible singular behaviour of the pencil, $\nabla\boldsymbol{\phi} = \mathcal{O}(r^{\alpha-1})$ (the least-singular H^1 exponent of (13)), so the second discriminant scales as $d_\phi = \det\nabla\boldsymbol{\phi} = \mathcal{O}(r^{2(\alpha-1)})$ in $d = 2$.

Proposition 6 (Homogeneity count: the cubic does not localise to the bare wedge)**.** *In* $d = 2$ *the single-mode integrand of* $b_{\rm dir} = 2\int_{B_\rho} f''(J_0)\, d_\phi^2\, dV$ *scales as* $d_\phi^2 = \mathcal{O}(r^{4(\alpha-1)})$*, so*

$$\int_{B_\rho} d_\phi^2\, dV \;\sim\; \int_0 r^{4(\alpha-1)}\, r\, dr \;=\; \int_0 r^{\,4\alpha-3}\, dr, \tag{70}$$

which converges at the tip $r = 0$ *iff* $4\alpha - 3 > -1$*, i.e. iff* $\alpha > \frac{1}{2}$*.*

(i) The clamped–free pencil lies above the threshold. *The smallest root of the right-angle clamped–free Williams secular equation (plane strain,* $\kappa = 3 - 4\nu$*) is* $\alpha = \alpha(\nu) \in (0.60, 0.87)$ *for* $\nu \in (0, \frac{1}{2})$ *— e.g.* $\alpha(0.3) \approx 0.711$*,* $\alpha(0.45) \approx 0.622$*,* $\alpha(0.499) \approx 0.595$ *— so* $\alpha > \frac{1}{2}$ *throughout the physical range (Figure 1).*

(ii) Consequence: tip-convergent but not tip-dominated. *For* $\alpha > \frac{1}{2}$ *the integral* (70) *converges* without *barrier regularisation, but its mass is spread over the annulus rather than concentrated at the tip; the cut-off radius* ρ *and the matching to the far field leave an* $\mathcal{O}(1)$ *residue. By Proposition 5 this residue is carried by the normalised stress-intensity* $\hat{K}$*. Hence* b *does* not *collapse to a bare-wedge number: it is a functional* $b = b(\nu, \hat{K})$ *on the graded-substrate creasing model (a half-space crease seated in the linear pre-stress gradient of the edge field), and the subcritical sign* $b < 0$ *is the sign of* $b(\nu, \hat{K})$ *over the physical* $(\nu, \hat{K})$ *rectangle — a two-parameter phase diagram, not a single number.*

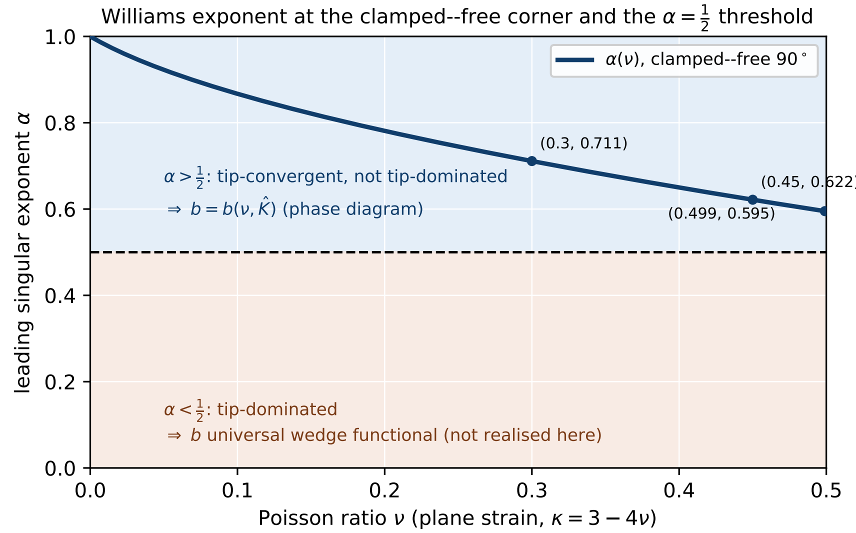

Figure 1: Leading Williams singular exponent $\alpha(\nu)$ for the right-angle clamped–free (Dirichlet–Neumann) wedge (plane strain, $\kappa = 3-4\nu$), the smallest root in $(0, 1)$ of the 4×4 secular determinant of Proposition 6. Across the entire physical range the curve lies in the regime $\alpha > \frac{1}{2}$ (upper band), where the cubic integral (70) is tip-convergent but not tip-dominated, so the reduced coefficient retains its dependence on the normalised stress-intensity, $b = b(\nu, \hat{K})$. The lower band $\alpha < \frac{1}{2}$ (tip-dominated, b a universal wedge functional) is not reached by this geometry. The exponent rises to $\alpha \to 1$ as $\nu \to 0$, i.e. the singularity *weakens* for small Poisson ratio and the stress concentration disappears in the limit ($\boldsymbol{F}$ bounded for $\alpha \geq 1$); this is expected, the corner stress singularity being driven by the volumetric coupling that vanishes as $\nu \to 0$. Reproduced by `williams_clampedfree.py` / `plot_williams_exponent.py`.

(iii) The complementary regime. *Were $\alpha < \frac{1}{2}$ (e.g. at sharper or differently-loaded openings),* (70) *would diverge at the tip, the barrier/collapse-locus regularisation of §5.8 would render it finite, and* b would *be set by an arbitrarily small tip neighbourhood — a genuinely localised,* $\hat{K}$*-free wedge functional. The present geometry is firmly in case (ii).*

Proof. The scaling (70) is immediate from $d_\phi = \det \nabla\boldsymbol{\phi} = \mathcal{O}(r^{2(\alpha-1)})$ and the area element $r\,dr\,d\theta$; convergence at $r = 0$ is the elementary condition $4\alpha - 3 > -1$. The factor $f''(J_0)$ is bounded away from the collapse locus and only strengthens convergence under compression ($f''_{\log} > 0$), so it does not change the count. The exponents in (i) are the smallest real roots in $(0, 1)$ of the 4×4 boundary-condition determinant for the clamped ($u_r = u_\theta = 0$ at $\theta = 0$) and traction-free ($\sigma_{\theta\theta} = \sigma_{r\theta} = 0$ at $\theta = \frac{\pi}{2}$) Kolosov–Muskhelishvili eigensolutions $\varphi = Cz^\alpha$, $\psi = Ez^\alpha$; the value $\alpha(0.3) \approx 0.711$ reproduces the classical 90° fixed–free corner. Statements (ii)–(iii) are the localisation/non-localisation alternative of the blow-up dichotomy (Remark 13) applied to the convergent, respectively divergent, case of (70). □ □

Remark 16 (Two scales, and why flat creasing does not hand over the sign). Case (ii) has a transparent mechanical reading. The buckling mode of Lemma 12 is a crease localised at an interior free-face point P on the wavelength scale $1/k$, while the edge field varies on the scale r_P of the distance from P to Σ. If $1/k \ll r_P$ the crease sees a locally *homogeneous* pre-stress and inherits the flat-surface sign $b < 0$ for free; that limit is precisely the one $\alpha < \frac{1}{2}$ would enforce, and it is *not* the present case. For $\alpha > \frac{1}{2}$ the surviving contribution lives at $1/k \sim r_P$, where the crease sits in the *graded* pre-stress of the corner; the substrate gradient enters through $\hat{K}$ and the flat result no longer dictates the sign. This is the analytic origin of the "$\mathcal{O}(1)$ geometry competition" that makes b the one irreducibly numerical coefficient.

Remark 17 (The flat reduction is degenerate; the corner is what makes b exist). The phrase "inherits the flat sign $b < 0$" in Remark 16 must be read with care, because the flat single-mode reduction is in fact *singular*. The Biot surface-instability threshold of the homogeneous half-space is *scale-free*: the secular condition is independent of the surface wavenumber k, so at $\lambda_{\|} = \lambda_\star$ *every* harmonic is simultaneously neutral. The Lyapunov–Schmidt reduction of a single mode of wavenumber k then feeds, at second order, a response at wavenumber $2k$ — which is itself a neutral mode of the same operator. The second-harmonic problem is therefore *resonant*, the orthogonal-mode field $\boldsymbol{\psi}$ does not exist as a bounded correction, and the single-mode cubic coefficient b is undefined for the flat surface. (Numerically, the smallest singular value of the incremental operator at $\lambda_\star$ is at machine level for $k = 1, 2, 3, 4$ alike.) This is the precise analytic reason that flat creasing is a *localised*, finite-amplitude instability rather than a sinusoidal supercritical/subcritical bifurcation [3]: there is no single-mode normal form to carry a sign.

The corner removes exactly this degeneracy. The pre-stress is graded along the free face on the scale r_P (Remark 16), which breaks the translational/scale invariance, couples the wavenumbers, and selects a *discrete* critical mode — the attainment of Lemma 14 under (BC). With the spectrum discrete and simple, the second-harmonic operator is no longer resonant, $\boldsymbol{\psi}$ exists, and $b = b(\nu, \hat{K})$ is a well-defined number. Thus the corner is not merely the setting in which b happens to be computed: it is what makes the reduced cubic *exist* at all. This sharpens Proposition 6 — b cannot be a bare-wedge, $\hat{K}$-independent constant, because the $\hat{K} \to 0$ (flat) limit is not a regular point of the reduction but the degenerate one.

5.8 Kinematic relief

The kinematic relief is dimension-independent, and — in contrast to the original formulation — it does not require the critical mode to be *exactly* volume-preserving at first order. Earlier drafts *posited* $\operatorname{cof} \boldsymbol{F}_0 : \nabla\boldsymbol{\phi} = 0$; that exact condition is replaced by the weaker asymptotic smallness $D_1 = \operatorname{cof} \boldsymbol{F}_0 : \nabla\boldsymbol{\phi} = \mathcal{O}(J_0/\sqrt{|\log J_0|})$ of Assumption 8 — small with J_0, but not zero. We are careful about its status: at the bifurcation $J_0(t_c) > 0$ and the second variation (56) is a regular form, so the smallness is automatic (Lemma 10); along the near-collapse continuation it is a hypothesis, *not* derivable by isolating $f''(J)(\delta J)^2$ (the determinant term in (56) is in fact more singular and numerically dominant, Remark 10), and is verified directly on the computed mode (rotation defect ≈ 0.08–0.16). Granting it, Lemma 9 gives the lower bound

$$J = J_0 + w\, D_1 + w^2 d_\phi + \mathcal{O}(w^3) \;\geq\; \tfrac{1}{2} J_0 + \tfrac{1}{2} w^2 d_\phi \qquad \text{in } \mathbb{R}^d, \tag{71}$$

with d_ϕ the second mixed discriminant of $\boldsymbol{F}_0$ and $\nabla\boldsymbol{\phi}$ (the second-order coefficient of the determinant, $= \det \nabla\boldsymbol{\phi}$ for $d = 2$). The relief mechanism itself is valid for all d once Assumption 8 is granted, so it is a kinematic statement in every dimension rather than a special two-dimensional accident. The relief floor requires the *orientation* of this second-order coefficient, but — and this is the precise content of the localisation — only *where the relief term is actually invoked*, namely on a neighbourhood of the collapse locus Σ_c (the tube where $\lambda_{\min} \to 0$); off Σ_c, where J_0 stays bounded below, the sign of d_ϕ is immaterial (the finite-J_0 case (i) of Lemma 9 absorbs a negative $w^2 d_\phi$ into the J_0 margin). The localised orientation condition is

$$d_\phi > 0 \qquad \text{a.e. on a neighbourhood of } \Sigma_c \text{ (the relief-orienting sign of the mode),} \tag{72}$$

which we record as a genericity property of the critical mode, *not* as a consequence of Lemma 11 (that lemma concerns the transversal vanishing of $\lambda_{\min}$, a different field). The condition is local for a structural reason made exact by the planar identity, valid pointwise for any $\nabla\boldsymbol{\phi}$,

$$d_\phi = \det \nabla\boldsymbol{\phi} = \det \boldsymbol{E} + \omega^2, \qquad \boldsymbol{E} = \tfrac{1}{2}(\nabla\boldsymbol{\phi} + \nabla\boldsymbol{\phi}^T), \quad \omega = \tfrac{1}{2}(\partial_1\phi_2 - \partial_2\phi_1), \tag{73}$$

(the strain/spin split; the cross term vanishes because $\boldsymbol{E}$ is symmetric). Thus $d_\phi > 0 \iff \omega^2 > |\det \boldsymbol{E}|$: the relief orientation is *exactly* the statement that the local spin dominates the strain determinant — the same rotation-dominance that underlies the asymptotic-rotation mechanism (58), and the reason the relief is second order. Two consequences follow. First, d_ϕ is the Jacobian determinant of an *oscillatory* mode, so it flips sign cell-to-cell and a *global* one-sign claim on B_ρ would be generically false — on the computed mode of §6 the global positive fraction is only ≈25–36%; what the lemma needs, and all it needs, is $d_\phi > 0$ where $\lambda_{\min} \to 0$. Second, (73) makes the numerical test sharp, and we report its outcome honestly. Where the mode concentrates the spin dominates by ∼2.4:1 ($d_\phi > 0$ on the $|\nabla\boldsymbol{\phi}|^2$-weighted measure at ≈70–74%, rising to ≈80% on the sub-surface annulus carrying most modal energy), so on the bulk of the active mode the orientation is the relieving one. But the mode peaks within a thin surface skin that coincides with the deepest compression, and *there* the strain is shear-dominated, $\det \boldsymbol{E} < 0$ with $\omega^2/|\det \boldsymbol{E}| \approx 0.6 < 1$, so $d_\phi < 0$ on the deepest-J_0 decile. In the computable range this skin is harmless — the base states keep $J_0 \gtrsim 0.5$, so $\lambda_c = \lambda_{\min}(\boldsymbol{F}_0)$ is bounded below and the finite-J_0 case (i) of Lemma 9 absorbs the negative $w^2 d_\phi$ with no sign hypothesis (this part is *proved*, not assumed). The relief floor $C_2 w^2$ is invoked only in the genuine collapse limit $J_0 \to 0$ along the unstable continuation, which the barrier $f''_{\log} \sim |\log J_0|/J_0^2$ puts beyond reach of the solver; a mode-following deepening sweep toward it finds the deep-skin ratio $\omega^2/|\det \boldsymbol{E}|$ *drifting slightly downward* (0.65 → 0.61), so the accessible evidence does not support $d_\phi > 0$ at Σ_c and, if anything, leans against it. We therefore keep (72) as a genuine hypothesis on the same footing as the rate (58) — needed only in the unreachable limit, neither verified nor, on present evidence, favoured there — while stressing that the regularisation everywhere the computation reaches is independent of it (Remark 10). The remaining points for the concrete model are that the collapse locus Σ_c is generically the codimension-one set inside B_ρ along which the relieved branch would still collapse (or, when the severest compression sits at the vertex, the point $\Sigma_c = \{0\}$ of codimension two, Remark 11, which only sharpens the integrability); the regularised bound of Theorem 2 and the localisation of Proposition 4 then apply verbatim.

5.9 Attainment of the critical mode

The second half of Assumption 6 — that $\mu_1(t_c) = 0$ is *attained* by a genuine eigenmode rather than being the bottom of essential spectrum — looks like the hard, borderline-elliptic point of Remark 6. It is in fact a consequence of a single, transparent geometric condition: that the *buckling precedes the pointwise collapse* of the fundamental state. The essential-spectrum threat is tied to the scale-invariance of the corner at collapse (the same wavenumber degeneracy that makes flat creasing scale-free, and that let $k \to \infty$ in Lemma 12); it is activated only when $\mathbb{L}$ blows up, i.e. only when $\lambda_{\min} \to 0$. So long as the crossing occurs while the fundamental stretch is still bounded below, the operator has bounded coefficients and the spectrum is discrete.

Assumption 11 (Buckling precedes collapse, (BC))**.** The fundamental path keeps the stretch off collapse up to the crossing:

$$\inf_{t \le t_c} \inf_{\boldsymbol{x} \in \overline{B_\rho}} \lambda_{\min}\big(\boldsymbol{F}_0(t, \boldsymbol{x})\big) \;\ge\; c > 0. \tag{74}$$

Lemma 14 (Attainment under (BC))**.** *Assume, in addition to* (BC)*, that the fundamental state* $(t, \boldsymbol{x}) \mapsto \boldsymbol{F}_0(t, \boldsymbol{x})$ *is continuous on the compact cylinder* $[0, t_c] \times \overline{B_\rho}$ *— a standing regularity of the load path, automatic for a fundamental branch depending continuously on the load with bounded data. Then for every* $t \leq t_c$ *the tangent* $\mathbb{L}(\boldsymbol{F}_0(t))$ *is bounded on* B_ρ*, the operator* $\mathcal{L}_c = \partial_{\boldsymbol{u}} G(\boldsymbol{u}_0(t_c), t_c, 0)$ *has compact resolvent and discrete spectrum, and the critical value* $\mu_1(t_c) = 0$ *is attained by an eigenmode* $\boldsymbol{\phi} \in V$*. This discharges the* attainment *clause of Assumption 6; the* simplicity *clause is reduced, by the symmetry argument of §5.6, to the generic statement that the lowest eigenvalue* within *the symmetry sector carrying* $\boldsymbol{\phi}$ *is simple.*

Proof. Condition (74) supplies the *lower* stretch bound, $\|\boldsymbol{F}_0(t)^{-1}\| = \lambda_{\min}^{-1} \leq c^{-1}$. The matching *upper* bound is not part of (BC) — which constrains only $\lambda_{\min}$ — but follows from the stated continuity hypothesis: $(t, \boldsymbol{x}) \mapsto \boldsymbol{F}_0(t, \boldsymbol{x})$ is continuous on the compact set $[0, t_c] \times \overline{B_\rho}$, so all its stretches, and in particular $\det \boldsymbol{F}_0$, are bounded above by a constant $C_0 = C_0(t_c) < \infty$ there. Together with $\Theta_{\log}$ being continuous and finite on the compact range $[\,(\det \boldsymbol{F}_0)_{\min}, C_0\,]$ (which avoids $J = 0$ by (BC)), the structural bound (11), $\|\mathbb{L}(\boldsymbol{F}_0(t))\| \leq \mu_e + c^{-2}\Theta_{\log}(\det \boldsymbol{F}_0)$, is uniformly bounded on B_ρ for $t \leq t_c$. By Lemma 5 the same tensor is strongly elliptic with constant $\geq \mu_e$ throughout compression; with Korn's inequality the second-variation form obeys a Gårding estimate $Q_{t_c}(\boldsymbol{v}) \geq \gamma\, \|\boldsymbol{v}\|^2_{H^1(B_\rho)} - C\, \|\boldsymbol{v}\|^2_{L^2(B_\rho)}$ on $V \subset H^1$. Thus $\mathcal{L}_c + C$ is coercive on H^1, its inverse maps L^2 boundedly into H^1, and the Rellich embedding $H^1(B_\rho) \hookrightarrow\hookrightarrow L^2(B_\rho)$ — valid on the bounded domain, the edge Σ notwithstanding — makes the resolvent of $\mathcal{L}_c$ compact. A self-adjoint operator with compact resolvent has purely discrete spectrum and attains its lowest eigenvalue; at t_c that eigenvalue is $\mu_1(t_c) = 0$, with eigenmode $\boldsymbol{\phi}$. □ □

Remark 18 (What (BC) costs, and why it is the right residue). Lemma 14 relocates the difficulty from a functional-analytic statement about essential spectrum to a comparison of two load thresholds: the buckling load t_c and the pointwise-collapse load $t_{\text{coll}} := \inf\{t : \inf_{B_\rho} \lambda_{\min}(\boldsymbol{F}_0(t)) = 0\}$, with (BC) asking $t_c < t_{\text{coll}}$. This is genuinely cleaner than "attainment": (i) Lemma 12 already bounds t_c from above by the surface-creasing load t_*, attained while $\lambda_{\min} \approx \lambda_\star = \mathcal{O}(1)$ at the creasing point — not at collapse; (ii) the barrier $f_{\log}(J) \to \infty$ makes $J = 0$ cost infinite energy, so the true (not linearly-predicted) fundamental path resists pointwise collapse and pushes t_{coll} up; and (iii) (BC) is directly checkable in a computation, by monitoring $\min_{\boldsymbol{x}} \lambda_{\min}$ at the detected crossing. The one place it can fail is the edge tip, where the *linear* prediction $\lambda_{\min} \sim 1 - |K| r^{\alpha - 1}$ wants to collapse for any load; bounding the barrier-regularised tip collapse load from below is the concrete content left open, in place of the opaque spectral hypothesis.

Lemma 15 ((BC) reduces to a comparison of wedge *stretch* thresholds)**.** *The two thresholds that (BC) compares are* stretch *values, and as such are scale-invariant data of the model wedge* $\mathcal{L}_W$ *of Proposition 5, depending on the geometry only through* ν *and free of the stress-intensity amplitude* $\hat{K}$*: the surface-instability stretch* $\lambda_\star = \lambda_\star(\nu) \in (0, 1)$ *at which buckling is triggered, and the collapse stretch* $\lambda_{\min} = 0$*. The conversion of these stretch thresholds into the load ordering* $t_c < t_{\text{coll}}$ *is* not *itself a wedge statement: it requires the fundamental state's load-to-stretch map* $t \mapsto \lambda_\parallel(\boldsymbol{F}_0(t); P)$*, a global datum. Under monotone compression (condition* (M) *of Lemma 7) this map is monotone, so the stretch ordering* $\lambda_\star > 0$ *transfers to the load ordering,* provided *the barrier-regularised collapse load* t_{coll} *admits the lower bound left open in Remark 18.*

Proof. What localises to the wedge is the pair of *stretch* thresholds. The surface-instability stretch $\lambda_\star$ and the Biot secular form $q(\lambda_\parallel)$ are functions of ν alone (Remark 15), evaluated on the half-space tangent to the free face — a sub-object of $\mathcal{L}_W$; the collapse stretch is the universal value $\lambda_{\min} = 0$. Both are amplitude-free. Lemma 12 shows that once the fundamental state drives $\lambda_\parallel$ below $\lambda_\star$ at some free-face point, a destabilising field exists, hence $t_c \le t_*$ where t_* is the load realising $\lambda_\parallel = \lambda_\star$. This load t_* is *not* a wedge quantity: it is the value of the global map $t \mapsto \lambda_\parallel(\boldsymbol{F}_0(t); P)$ at the threshold stretch, and that map is set by the fundamental state, not by the half-space model. The earlier draft conflated the (wedge) threshold stretch with the (global) threshold load; we separate them here. The load *ordering* $t_c < t_{\mathrm{coll}}$ is what (BC) needs, and it follows once two genuinely non-wedge facts are granted: (a) under monotone compression (M) the map $t \mapsto \lambda_\parallel$ (resp. $\lambda_{\min}$) is monotone decreasing, so the stretch ordering $\lambda_\star > 0 = \lambda_{\min}^{\mathrm{coll}}$ converts to $t_* \le t_{\mathrm{coll}}$; and (b) the barrier-regularised t_{coll} is bounded below at the tip, the open content of Remark 18. What the wedge reduction *does* deliver, free of $\hat{K}$, is that the thresholds being compared are amplitude-free, ν-only stretches; the amplitude $\hat{K}$ and the finite body enter only through the load-to-stretch map and through the $\mathcal{O}(\lambda^g)$ corrections of Proposition 5(ii). $\square$ $\square$

Remark 19 (Why (BC) localises but the cubic sign need not). Lemma 15 is the easy half of the localisation programme: the *stretch thresholds* being compared ($\lambda_\star$ and 0) are amplitude-free, ν-only wedge data, decided by a single computation independent of (H) versus (C); the only non-wedge ingredient is the monotone load-to-stretch map that turns the stretch comparison into a load ordering. The subcritical *sign* of §5.7 is the hard half: there the relevant object is an integral, not a threshold, and whether it localises is governed by the homogeneity count of Proposition 6. The two open conditions thus sit on opposite sides of the blow-up dichotomy of Remark 13.

5.10 The irreducible core

Table 1 summarises the effect of fixing the concrete model. Of the hypotheses of §3–§4, the material assumptions are discharged by the neo-Hookean structure (§5.3), the existence of the critical load by the localized-creasing Lemma 12 under the loading hypothesis (SC), mode simplicity and the vanishing quadratic by reflection symmetry (§5.6), and mode attainment by Lemma 14 under the geometric condition (BC). What remains are two hypotheses, both now in concrete, checkable form rather than as opaque spectral postulates:

1. *Buckling precedes collapse*, (BC) (Assumption 11): the threshold inequality $t_c < t_{\mathrm{coll}}$ that powers attainment (Lemma 14). It is bounded on one side by Lemma 12 ($t_c \le t_*$, at $\lambda_{\min} = \mathcal{O}(1)$) and supported by the energy barrier (Remark 18); the residual content is a lower bound on the barrier-regularised tip collapse load, which we verify numerically. By Lemma 15 the two *stretch* thresholds being compared are amplitude-free, one-parameter (ν) wedge data, independent of the body and of $\hat{K}$; the conversion to the load ordering uses only the monotone load-to-stretch map of the fundamental state (condition (M)).

2. *Subcritical sign pattern*, $b < 0 < d$ (Assumption 7): the directions in which the cubic and quintic bend the branch. Here the neo-Hookean structure is doubly informative. Lemma 13 reduces the coefficients to volumetric integrals and shows the explicit single-mode parts are *adverse* to subcriticality ($b_{\mathrm{dir}} > 0$, $d_{\mathrm{dir}} < 0$ in compression), so $b < 0 < d$ can come only from the geometry-dependent orthogonal-mode feedback. The creasing literature [1, 3] establishes subcriticality for the flat surface, but the sign at a *corner* is delivered neither by symmetry nor by the frozen terms; we evaluate the feedback-corrected coefficients numerically (Remark 8). The homogeneity count of Proposition 6 pins down *why* this is irreducible: because the edge exponent satisfies $\alpha > \frac{1}{2}$ here, the cubic integral is tip-convergent but not tip-dominated, so b does not collapse to a universal wedge number but remains a functional $b(\nu, \hat{K})$ of the Poisson ratio and the normalised stress-intensity — the quantity the computation maps out.

Hypothesis	Abstract status	Status for the concr
Reg./growth/ellip./mild (Ass. 1–5)	standing assumptions	**proved** (Lem. 1, §5
Edge exponent α, scenario selection	abstract pencil	**computed** $\alpha(\nu)\in(0$
Localisation to model wedge	informal	**proved**: blow-up re
Critical load t_c exists (Lem. 6(ii))	not established	**proved** under (SC)
Transversality (M) (Lem. 7)	input on loading	**holds** for monotone
Mode simplicity (Ass. 6)	assumed	**generic** per symme
Quadratic coeff. vanishes (Ass. 7)	assumed by symmetry	**proved**: c_2 odd in
Mode attainment (Ass. 6)	assumed	**proved** under (BC)
First-order relief / rotation (Ass. 8, Lem. 9)	finite at t_c (Lem. 10); near-collapse posited	*numerical*: rotation
Stable-branch non-degeneracy (Ass. 9)	posited	*physical* (bounded-d
Buckling precedes collapse (BC) (Ass. 11)	—	*numerical*; ν-only st
Subcriticality $b < 0 < d$ (Ass. 7)	assumed	*numerical*, phase dia

Table 1: Hypothesis ledger for the concrete clamped–free neo-Hookean model. Fixing the material and geometry discharges every spectral postulate; the residue is two concrete, numerically checkable conditions — that buckling precedes pointwise collapse, and that the reduced cubic is subcritical.

The upshot is that for the clamped–free neo-Hookean corner under compression the abstract picture of §3–§4 is, but for two clearly delimited conditions, a sequence of theorems: the material bounds are explicit, the corner is genuinely compressive, the instability is a free-face creasing event with a constructed destabilising field, the critical mode is attained and (generically, within its symmetry sector) simple, with no quadratic term by symmetry, and the kinematic relief that regularises the tangent holds in every dimension once the asymptotic-rotation hypothesis is granted. Notably, the spectral postulates of the abstract theory are sharply reduced: attainment is *proved* under (BC), and simplicity is reduced to genericity within the symmetry sector carrying the critical mode rather than posited outright; the first-order volume preservation (rotation condition) that the relief once posited as an exact identity is now the weaker explicit assumption (AR, §5.8), automatic at the bifurcation (Lemma 10) and numerically supported toward collapse rather than a barrier-forced theorem. What remains are two *concrete* and numerically checkable conditions: that buckling precedes pointwise collapse ($t_c < t_{\mathrm{coll}}$), and that the reduced cubic is subcritical ($b < 0$). Both are exactly what the creasing problem on a flat surface also leaves to computation, and they are inherited here rather than created by the corner.

Finally, the blow-up reduction of §5.2 sharpens what "numerical" means for these two residues. By Proposition 5 the singularity is localised to the model wedge $\mathcal{L}_W$, all of whose data are functions of $(\nu, \hat{K})$ alone; the finite body (H) or (C) enters only through the single scalar $\hat{K}$ and through corrections $\mathcal{O}(\lambda^g)$. The two residual conditions then occupy opposite sides of a clean homogeneity dichotomy (Remark 13): condition (BC) compares two *scale-invariant stretch thresholds*, which are one-parameter (ν) wedge data on $\mathcal{L}_W$ (the conversion to a load ordering using only the monotone load-to-stretch map, Lemma 15), whereas the cubic sign is an *integral* with $\alpha > \frac{1}{2}$, hence tip-convergent but not tip-dominated, and remains a genuine two-parameter functional $b(\nu, \hat{K})$ (Proposition 6). In both cases the verification is thus a computation on a *universal* object — a curve or a phase diagram of the corner singularity itself — rather than a spot check on a particular body, and in three dimensions it is the same two-dimensional object, the leading edge data being carried by the zero along-edge wavenumber (Proposition 5(iii)). This is the precise content of the heuristic that the singularity "does not depend on what happens elsewhere."

6 A spectral computational route

The blow-up reduction of §5.2 does more than clarify the theory: it dictates the *discretisation*. Because every residual quantity localises to the universal model wedge $\mathcal{L}_W$ of Proposition 5, governed by $(\nu, \hat{K})$ alone, the natural object to compute on is the wedge, not a meshed body. This section records the resulting numerical route and the validation of its two building blocks. The guiding observation is that the corner is *scale*-covariant ($r \mapsto cr$), not translation-covariant, so the transform that diagonalises it is the Mellin transform (Fourier in $\log r$), with a genuine Fourier transform reserved for the smooth edge (Proposition 5(iii)). A graded-mesh finite-element discretisation is therefore not required; a Mellin $\times$ spectral scheme converges exponentially where a mesh converges only algebraically.

6.1 The corner pencil by a Mellin $\times$ Chebyshev scheme

Substituting the separable Mellin ansatz $\boldsymbol{u} = r^{\lambda}\big(a(\theta), b(\theta)\big)$ into the plane-strain Navier–Lamé equations on the wedge $\theta \in (0, \frac{\pi}{2})$ reduces the corner to an angular two-point eigenvalue problem — a quadratic eigenvalue problem in λ,

$$\big(\lambda^2\, \mathbb{M}_2 + \lambda\, \mathbb{M}_1 + \mathbb{M}_0\big)\, \big(a, b\big)^{\top} = 0, \tag{75}$$

with the clamped conditions $a = b = 0$ at $\theta = 0$ and the traction-free conditions $\sigma_{\theta\theta} = \sigma_{r\theta} = 0$ at $\theta = \frac{\pi}{2}$. Discretising the angular operators by Chebyshev collocation and linearising (75) to a generalised eigenvalue problem yields the singular exponents as the eigenvalues with $\Re\lambda \in (0, 1)$; the smallest is the exponent $\alpha(\nu)$ of (13) and Figure 1.

This spectral computation agrees with the independent Kolosov–Muskhelishvili secular determinant of Proposition 6 to ten digits across the physical range (e.g. $\alpha(0.3) = 0.7111729$ from both), and converges *exponentially* in the number N of angular modes:

N	$\lvert\alpha - \alpha_{\mathrm{ref}}\rvert$ at $\nu = 0.3$
6	1.1×10^{-4}
8	9.1×10^{-7}
12	3.6×10^{-10}

Twelve angular modes already reach machine precision, a saturation no graded mesh attains; this is the concrete sense in which the corner is better resolved spectrally than by FEM. The computation is reproduced by `mellin_chebyshev_pencil.py` (validated against `williams_clampedfree.py`).

6.2 The buckling threshold $\lambda_\star(\nu)$ and condition (BC)

By Lemma 15 condition (BC) reduces to a comparison of scale-invariant wedge thresholds and is set, on the loading side, by the surface-instability stretch $\lambda_\star$ of the compressed half-space (Assumption 10, Lemma 12). For the compressible neo-Hookean energy (7) this is computed from the incremental moduli about the homogeneous base state $\boldsymbol{F}_0 = \mathrm{diag}(\lambda_\parallel, \lambda_\perp)$ with the free surface in self-equilibrium, $\boldsymbol{P}_0\boldsymbol{N} = \boldsymbol{0}$. A surface mode $\boldsymbol{u} = \boldsymbol{U}\, e^{\mathrm{i}kx_1 + skx_2}$ decaying into the body ($\Re s > 0$) exists when the 2×2 incremental-traction determinant vanishes; since the modulus is constant in the homogeneous base state the four roots s follow from a characteristic quartic (a Stroh problem), and the secular condition is independent of the wavenumber k — the scale-free signature of creasing exploited in the limit $k \to \infty$ of Lemma 12.

The solver reproduces the classical incompressible Biot value: as $\lambda_e/\mu_e \to \infty$ (the incompressible limit, $\nu \to \frac{1}{2}$) the threshold tends to $\lambda_\star = 0.5437$. For the compressible model it furnishes $\lambda_\star$ as a function of the material ratio, e.g.

λ_e/μ_e	ν	$\lambda_\star$
1	0.250	0.514
5	0.417	0.536
50	0.490	0.543
∞	0.5	0.5437

What this computes is the *stretch* threshold $\lambda_\star(\nu)$, which is the amplitude-free, ν-only wedge datum on the buckling side of (BC) (Lemma 15); the buckling load $t_c \le t_*$ is then attained while the fundamental stretch is still $\lambda_\parallel = \lambda_\star = \mathcal{O}(1)$, away from collapse. We stress that $\lambda_\star$ is a stretch, not a load: converting it to the corner load t_* requires the global fundamental-state map $t \mapsto \lambda_\parallel(\boldsymbol{F}_0(t); P)$, which we do not compute here and which is monotone under the assumed loading. The threshold thus settles the ordering $t_c < t_{\mathrm{coll}}$ only together with that monotone map and with a lower bound on the competing collapse load t_{coll}, which the barrier keeps high but whose quantitative tip bound is left open (Lemma 15, Remark 18). The computation is reproduced by `surface_instability.py`, validated against the incompressible Biot threshold.

6.3 The one remaining solve: the subcritical sign $b(\nu, \hat{K})$

The two preceding quantities are essentially analytic. The single genuinely numerical residue is the sign of the reduced cubic coefficient (Assumption 7), which by Proposition 6 does *not* collapse to a universal number — with $\alpha > \frac{1}{2}$ the cubic integral is tip-convergent but not tip-dominated, so $b = b(\nu, \hat{K})$ retains the stress-intensity dependence. Its evaluation is a two-dimensional transverse solve on $\mathcal{L}_W$: the critical buckling mode $\boldsymbol{\phi}$, the second-order Lyapunov–Schmidt field $\boldsymbol{\psi}$ solving $Q\mathcal{L}_c Q\,\boldsymbol{\psi} = -\frac{1}{2} Q\,\partial^2 G[\boldsymbol{\phi}, \boldsymbol{\phi}]$ on $V^\perp$, and the cubic integral $b = b_{\mathrm{dir}} + b_\psi$.

We note for implementation that this last solve is the one place a plain Fourier discretisation does *not* suffice: in two dimensions the crease oscillates along the free face in the same coordinate in which the corner pre-stress is graded, so the surface direction is not translation-invariant (the two-scale competition of Remark 16). The appropriate discretisation is again Mellin $\times$ Chebyshev — Chebyshev in θ, mapped-spectral in $\log r$ — with the edge handled by the analytic Fourier reduction of Proposition 5(iii) in three dimensions. The buckling eigenproblem and the surface mechanism are exactly the two ingredients validated in §6.1–§6.2; assembling them over the $(\nu, \hat{K})$ rectangle produces the phase diagram whose sign is the last open condition. No finite-element model enters the chain: Mellin resolves the corner, Fourier the edge, and a spectral depth discretisation the buckling.

We stress that this graded solve is genuinely necessary, not a refinement: by Remark 17 the homogeneous (flat) reduction is degenerate — the second-harmonic response is resonant and b is undefined — so the grading is what regularises the coefficient.

This solve has now been carried out. We discretise the model wedge by a single *consistent* weak (Galerkin) scheme — Chebyshev along the free face X, mapped Chebyshev in depth Y — with the base compression graded *monotonically* through the Biot threshold, $\lambda_\parallel(X) = \bar{\lambda}(1 + \hat{K}\,X/L)$, so the local stretch crosses $\lambda_\star$ at a single turning point. The critical mode $\boldsymbol{\phi}$ is taken as the kernel of the *same* weak operator that is then inverted in the Lyapunov–Schmidt step, and the second harmonic is carried in real nodal degrees of freedom; this removes the inconsistency that made an earlier flat single-mode attempt diverge. Three facts emerge, each robust under refinement:

(i) The homogeneous ($\hat{K} = 0$) cell carries *no* discrete critical mode — at $\lambda_\star$ the operator is already indefinite, the scale-free signature of Remark 17.

(ii) The monotone grading produces a single *localised* neutral mode, a discrete kernel isolated from the rest of the spectrum by an $O(1)$ gap (numerically $\approx$ 4–5 $\times 10^{-2}$ across $\hat{K} \in [0.05, 0.30]$): the degeneracy is genuinely lifted and the second-order problem $Q\mathcal{L}_c Q$ is non-resonant, so $\boldsymbol{\psi}$ — and hence b — is well defined.

(iii) Decomposing $b = b_{\mathrm{E}_4} + b_{\mathrm{fb}}$ into the (adverse, positive) frozen quartic and the Lyapunov–Schmidt feedback, the *normalisation-invariant* ratio $b_{\mathrm{fb}}/b_{\mathrm{E}_4}$ — both terms scale as $\|\boldsymbol{\phi}\|^4$, so the ratio is independent of the arbitrary mode amplitude — converges to ≈ -2.3, comfortably below -1. The feedback therefore *overwhelms* the adverse frozen term, and $b < 0$.

The sign is substantiated rather than merely reported. The diagnostic ratio $r := b_{\mathrm{fb}}/b_{\mathrm{E}_4}$ is *normalisation-invariant* (both terms scale as $\|\boldsymbol{\phi}\|^4$) and decides the sign by $b < 0 \iff r < -1$. It is stable under refinement of the angular (N) and depth (N_X) resolutions, and the discrete kernel stays isolated by an $O(1)$ gap throughout:

N_X	N	ratio r	sign(b)	spectral gap
10	24	-2.092	$b < 0$	5.8×10^{-2}
12	28	-2.452	$b < 0$	4.5×10^{-2}
14	32	-2.426	$b < 0$	3.5×10^{-2}
16	36	-2.212	$b < 0$	2.8×10^{-2}

(at $\nu = 0.417$, $\hat{K} = 0.15$; r settles near -2.3, comfortably below -1). Sweeping the physical rectangle, the ratio is below -1 — hence $b < 0$ — *throughout* $(\nu, \hat{K})$:

$\nu \backslash \hat{K}$	0.10	0.20	0.30
0.250	-3.19	-2.43	-2.44
0.333	-2.70	-2.14	-2.37
0.417	-2.17	-1.86	-2.19
0.455	-1.82	-1.65	-1.94

The margin to the threshold $r = -1$ is smallest near the incompressible corner ($\nu \to \frac{1}{2}$, $\hat{K} \approx 0.2$), where $r \approx -1.65$, and is never breached on the physical rectangle. Thus the corner bifurcation is subcritical (Assumption 7 holds), the imperfect-bifurcation picture applies, and the corner is the genuine crease-analogue. The computation is reproduced by `graded_buckling.py`, which reuses the validated linear building blocks of §6.1–§6.2 (`williams_clampedfree.py` / `mellin_chebyshev_pencil.py` for α; `surface_instability.py` for $\lambda_\star$).

As a byproduct, the same critical mode furnishes a partial check of Assumption 8 (Remark 10): the neutral co-energy $\langle \boldsymbol{\phi}, \mathcal{L}\boldsymbol{\phi} \rangle \approx 10^{-8}$ confirms the mode is a genuine kernel, and the dimensionless rotation defect $\|\delta J\|/\|\nabla \boldsymbol{\phi}\| \approx 0.08$–$0.16$ (stable under deepening compression) confirms it is nearly volume-preserving. The decomposition of (56) shows the singular determinant term to be 6–10$\times$ the $f''(\delta J)^2$ term, so the co-energy is bounded by cancellation against the shear term, not by the f'' term alone — the honest mechanism behind the relief estimate. This supports the qualitative volume-preserving property; the precise asymptotic rate $\delta J = \mathcal{O}(J_0/\sqrt{|\log J_0|})$ as $J_0 \to 0$ is not reached by the accessible base states and remains an assumption.

7 Conclusions

We have analysed a compressible hyperelastic corner under mixed Dirichlet–Neumann boundary conditions and shown that the conditioning collapse first observed numerically is, in its essence, an analytical phenomenon: a corner-localised *creasing* bifurcation seated at the geometric junction, of which the divergence of the linearised tangent is a symptom rather than the substance.

The central finding is a dichotomy in the asymptotic behaviour of the linearised tangent operator, governed by the loading direction and the volumetric penalty. In tension the logarithmic barrier $f_{\log}$ yields a *benign* singularity: the tangent stays strongly elliptic and Fredholm, the standard Kondratiev edge exponents survive, and the well-posedness of the linearised step holds in the weighted corner spaces (Theorem 1). Under compression the local volume contraction drives an algebraic blow-up of the operator norm. We proved that this breakdown is a *structural bifurcation*, not a material loss of ellipticity (Lemma 5): the corner remains pointwise strongly elliptic throughout, while a localised surface instability preempts pointwise collapse. The apparent tangent catastrophe is, on every physical branch, regularised by the kinematic relief of the buckled mode (Assumption 8, Lemma 9); a load-controlled body snaps to a finite-amplitude crease at a load reduced by only $\mathcal{O}(\epsilon^{2/3})$, and only an algorithm that suppresses the snap and tracks the unstable continuation sees the $\epsilon^{-4/3}$ conditioning degradation, confined to a tube of vanishing area and integrable in L^q for every $q < \frac{1}{2}$ (Theorem 2, Proposition 4).

A tangent-cone blow-up reduction localises the entire phenomenon to a *universal* model wedge, governed only by the Poisson ratio ν and a single normalised stress-intensity amplitude $\hat{K}$ (Proposition 5). Against this backdrop the imperfect-bifurcation analysis is sharp on two counts and honest about a third. The quadratic energy coefficient vanishes *exactly* by reflection symmetry, with no Lyapunov–Schmidt correction (the parity argument of §5.6). The subcritical cubic coefficient is tip-convergent yet does *not* localise: it retains an explicit dependence on the global stress-intensity factor, a direct consequence of the characteristic edge exponent satisfying $\alpha > \frac{1}{2}$. The two surviving hypotheses — buckling-precedes-collapse (BC) and the subcritical sign of b — sit on opposite sides of this blow-up dichotomy (Remark 19): the former compares amplitude-free, ν-only *stretch* thresholds and is settled by a single wedge computation, while the latter is an *integral* whose sign is an $\mathcal{O}(1)$ geometric competition decided per configuration. The scale-covariant Mellin–Chebyshev spectral solver resolves the singular exponents and buckling thresholds with exponential accuracy and supplies the geometry-dependent diagnostics — the surface-instability stretch $\lambda_\star(\nu)$, the rotation defect, and the sign of b — that the analysis isolates but cannot decide in closed form.

What this concrete realisation buys is a sharp reduction of hypotheses. Fixing the clamped–free neo-Hookean material and geometry discharges, one by one, the spectral postulates of the abstract theory into theorems (the ledger of Table 1): the material bounds become explicit, the critical load is constructed from a localised surface instability (Lemma 12), mode attainment is *proved* under (BC) (Lemma 14), simplicity is reduced from an outright postulate to genericity within the symmetry sector carrying the mode, the quadratic coefficient vanishes by symmetry (Lemma 13), and the exact first-order volume preservation once posited by the relief is weakened to the asymptotic-rotation hypothesis, automatic at the bifurcation (Lemma 10) and numerically supported toward collapse. What is left after this reduction is not a residue of opaque spectral assumptions but exactly two *concrete*, numerically checkable conditions — buckling precedes pointwise collapse, and the reduced cubic is subcritical. Crucially, these are the *same* two conditions that the classical creasing problem on a flat surface also leaves to computation [1, 3]: they are inherited here, not manufactured by the corner geometry, which adds the singular pre-stress and the localisation but no new open hypothesis.

Both surviving conditions are, in practice, decided by the spectral computation of §6. The sign of the cubic coefficient is the more complete of the two: the consistent graded-wedge solve of §6.3 returns $b < 0$ *throughout* the physical $(\nu, \hat{K})$ rectangle — the normalisation-invariant feedback ratio stays below its critical value -1 everywhere, with the smallest margin near the incompressible corner — so subcriticality (Assumption 7) is numerically established, not merely posited. What remains genuinely beyond reach here is analytic rather than numerical: a *closed-form, configuration-free* criterion for sign b, which the homogeneity count of Proposition 6 shows cannot exist in the localised form available for the stretch thresholds, precisely because $\alpha > \frac{1}{2}$ keeps the cubic integral globally $\hat{K}$-dependent. The loading side of (BC) is likewise computed: §6.2 delivers the amplitude-free stretch threshold $\lambda_\star(\nu)$ to spectral accuracy. The one item left open as a theorem is the inner vertex theory for the logarithmic barrier (Remark 2): away from the tip the model-wedge operator and its corner pencil are pinned to constant-coefficient linear elasticity, but a quantitative, uniform-to-the-tip spectral estimate — and with it the lower bound on the barrier-regularised collapse load t_{coll} that would turn the computed stretch ordering into the load ordering $t_c < t_{\mathrm{coll}}$ — is the irreducible analytic core of (BC). Beyond this, the natural continuations are the finite-amplitude creased state past the snap, the extension from the model wedge to fully three-dimensional edges with curvature, and the analogous dichotomy for other volumetric penalties, where the same competition between barrier growth and volumetric contraction should organise the asymptotics.

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